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<i>H01L 29/786</i>	(2006.01)
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<i>H10D 30/67</i>	(2025.01)
<i>H10D 62/10</i>	(2025.01)
<i>H10D 64/01</i>	(2025.01)
<i>H10D 64/23</i>	(2025.01)

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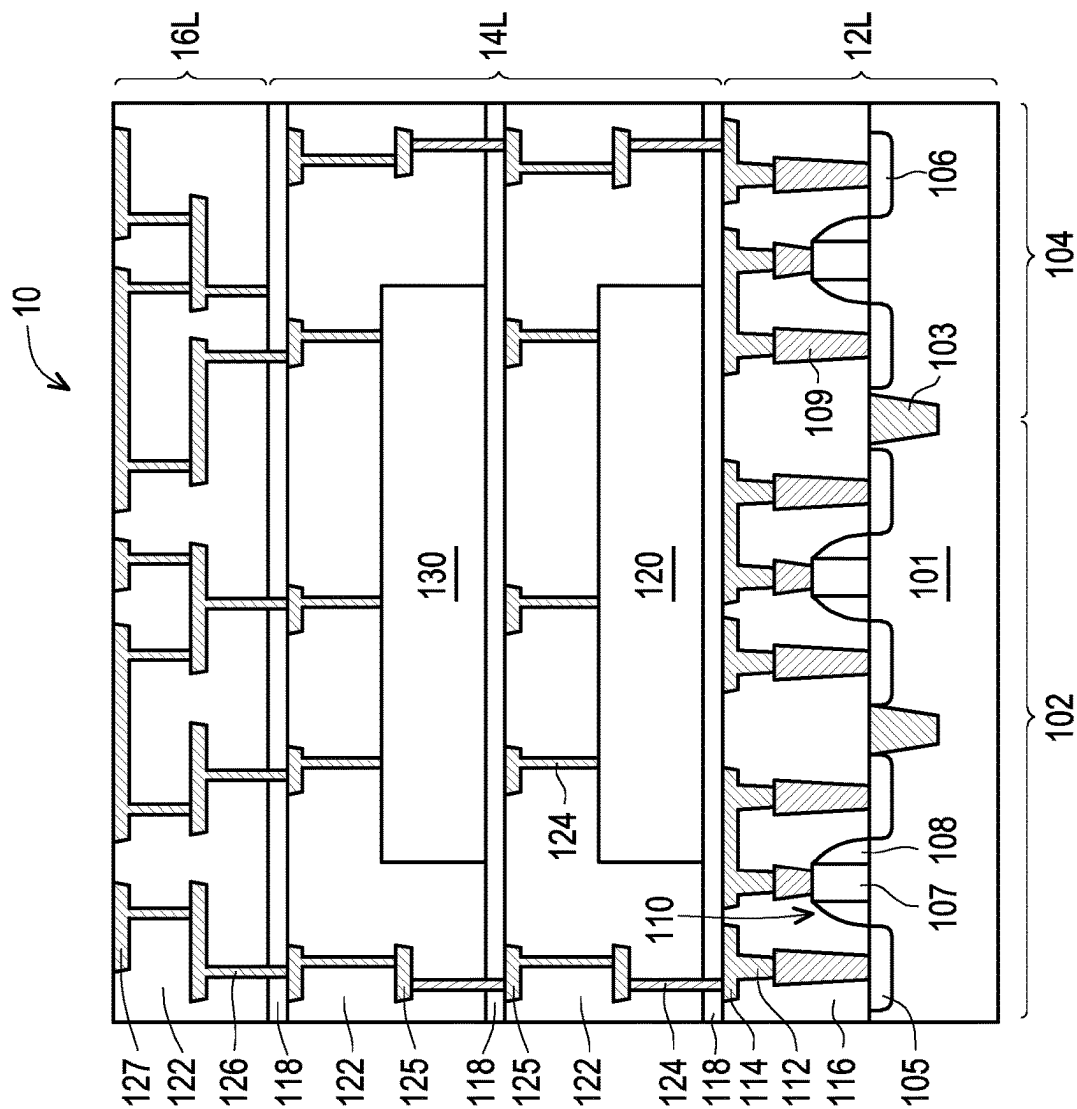


FIG. 1

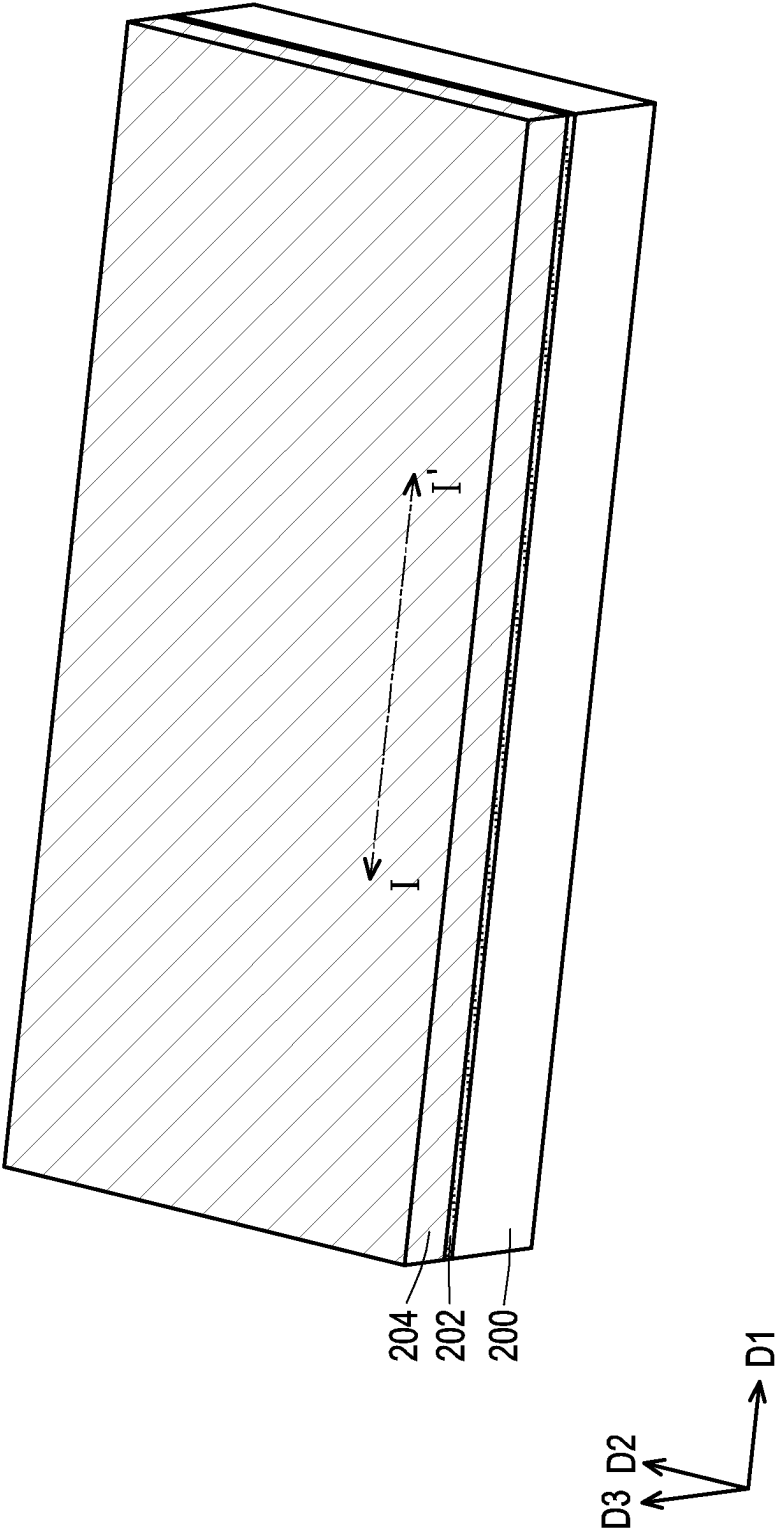


FIG. 2A

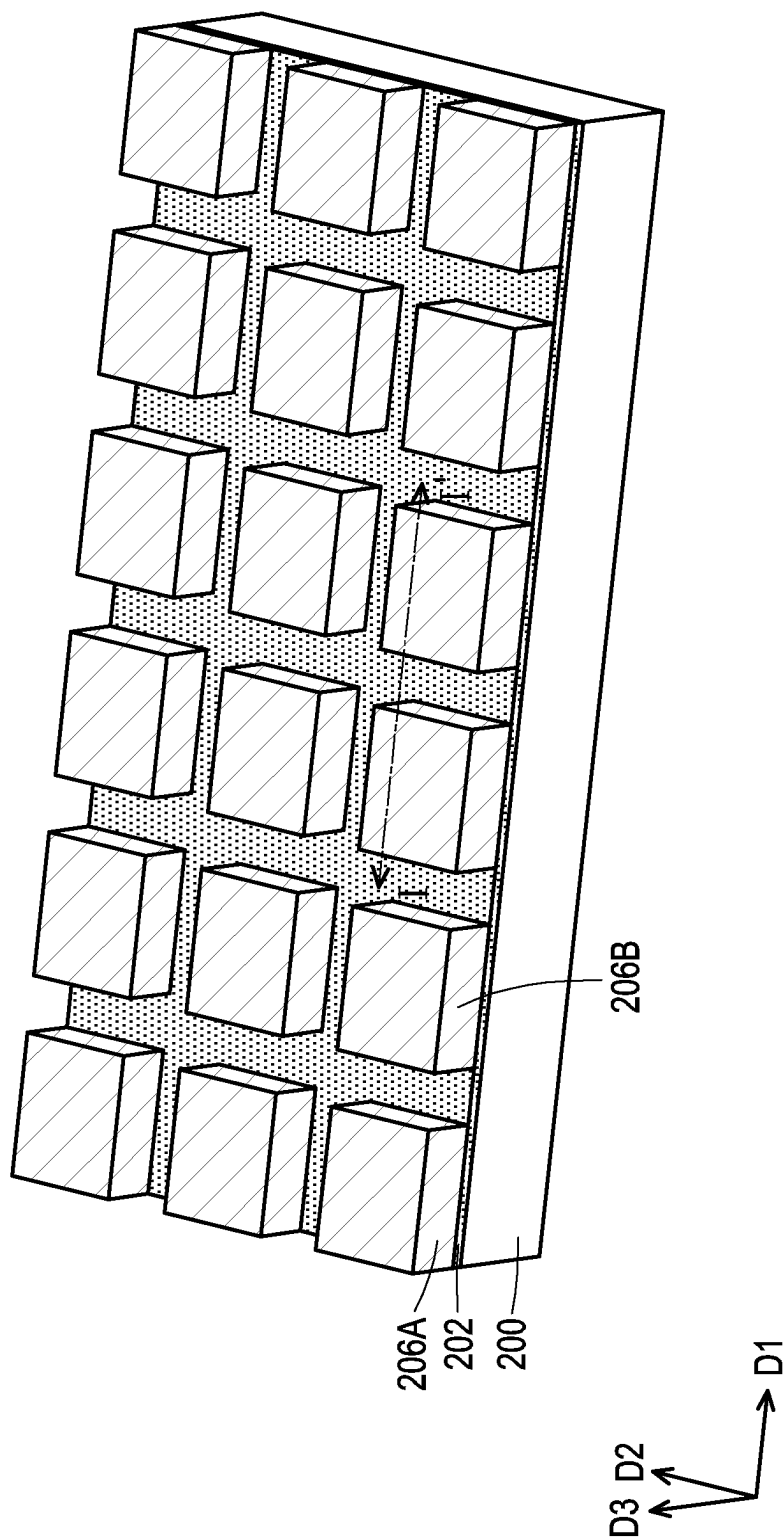


FIG. 2B

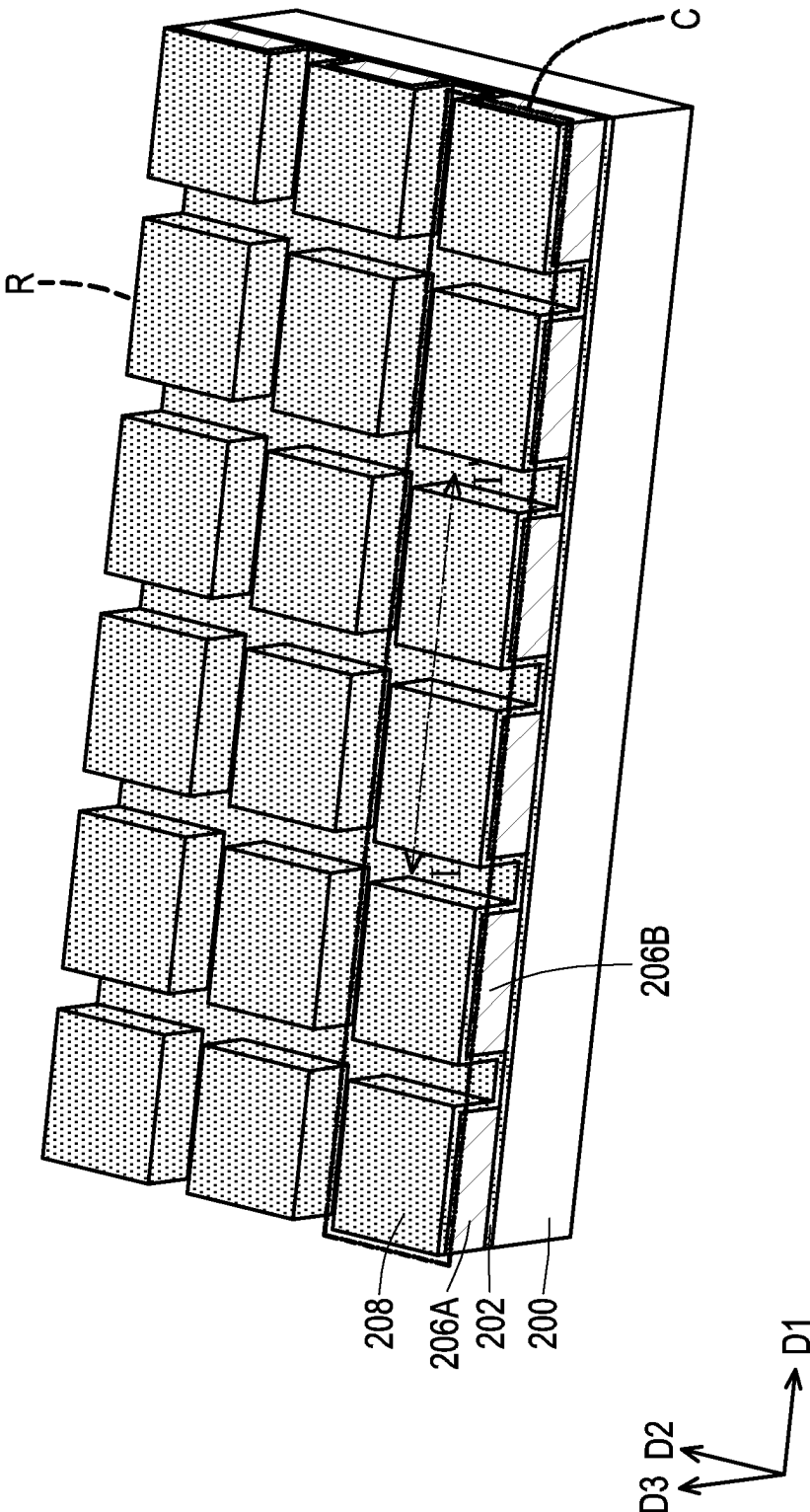


FIG. 2C

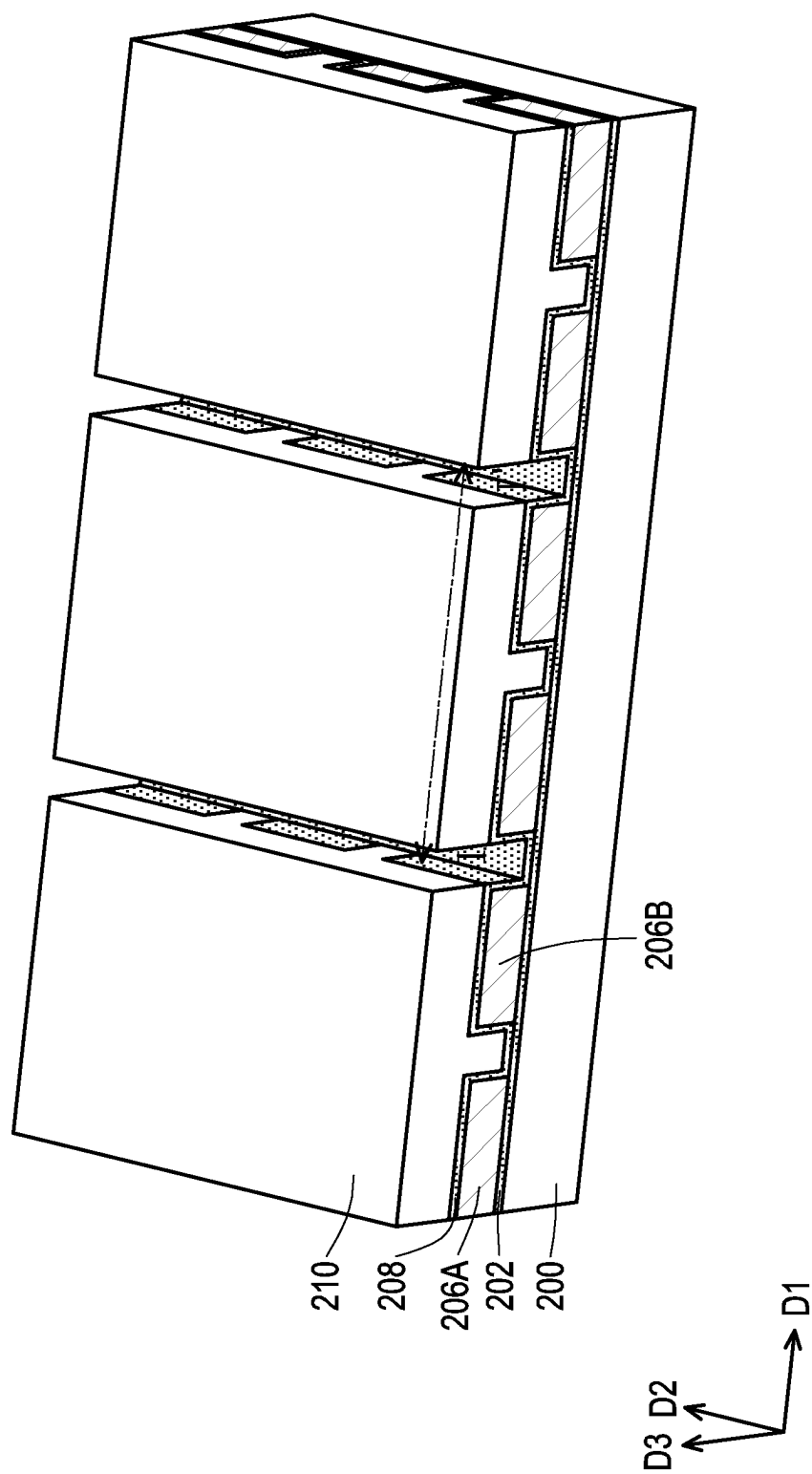


FIG. 2D

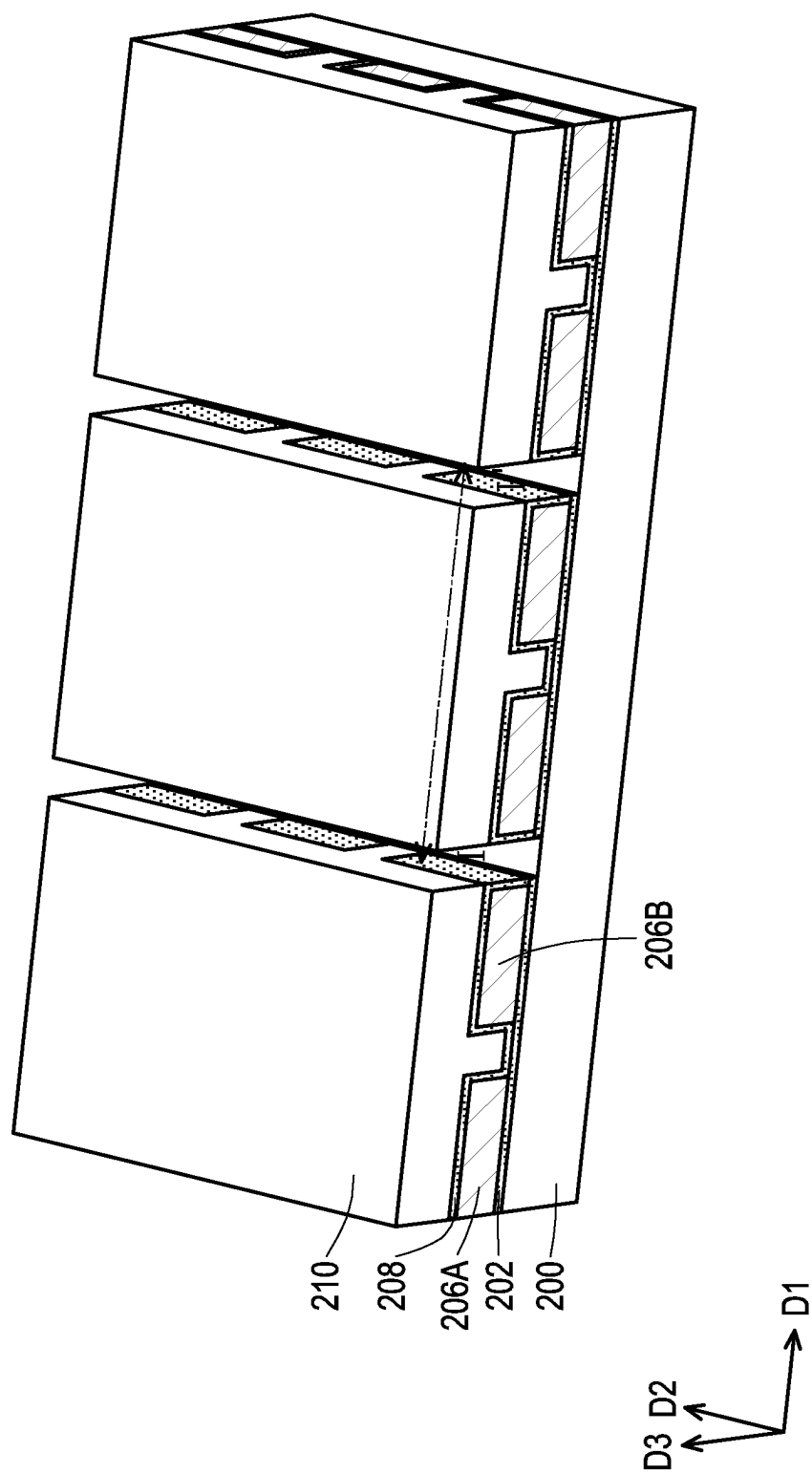


FIG. 2E

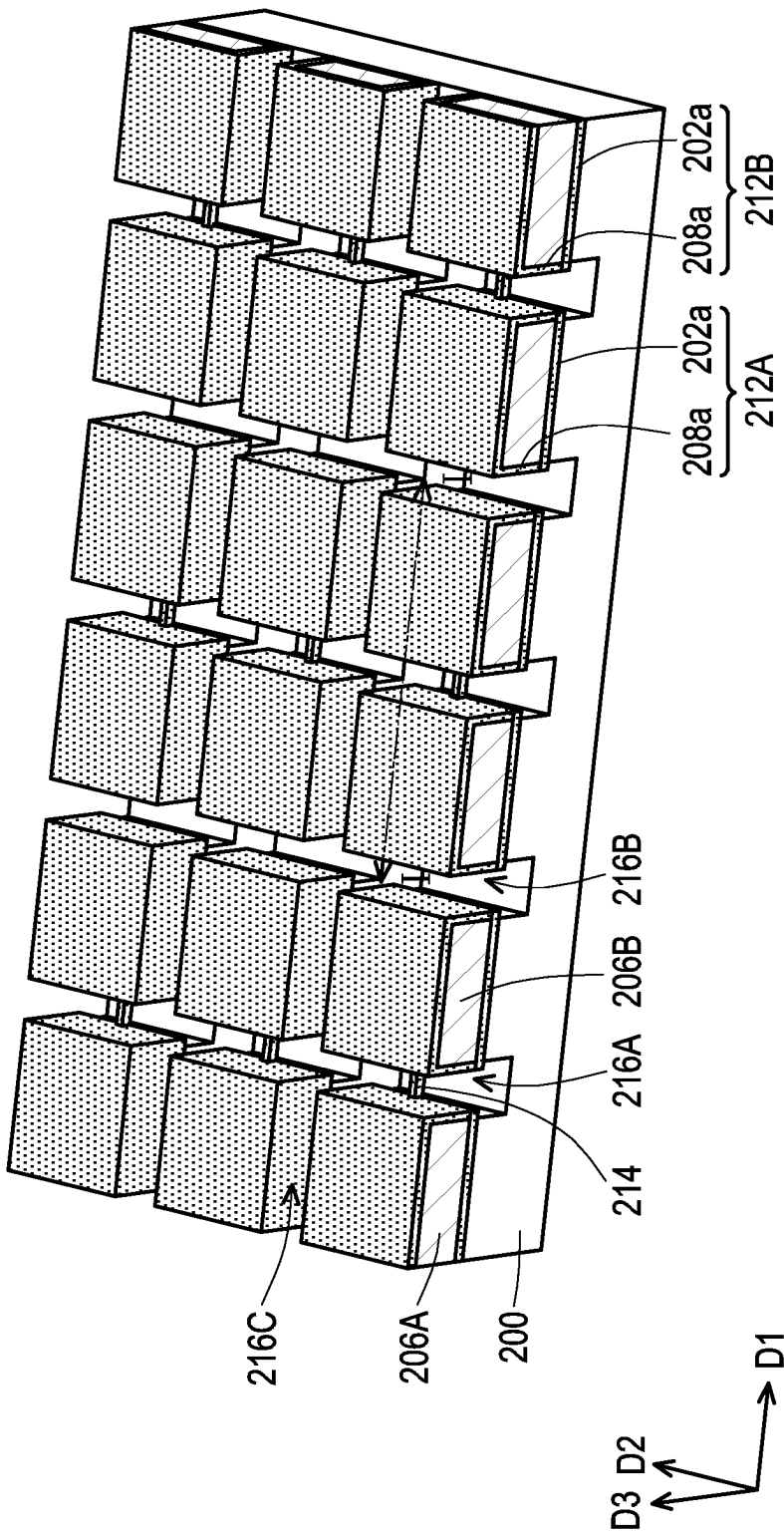


FIG. 2F

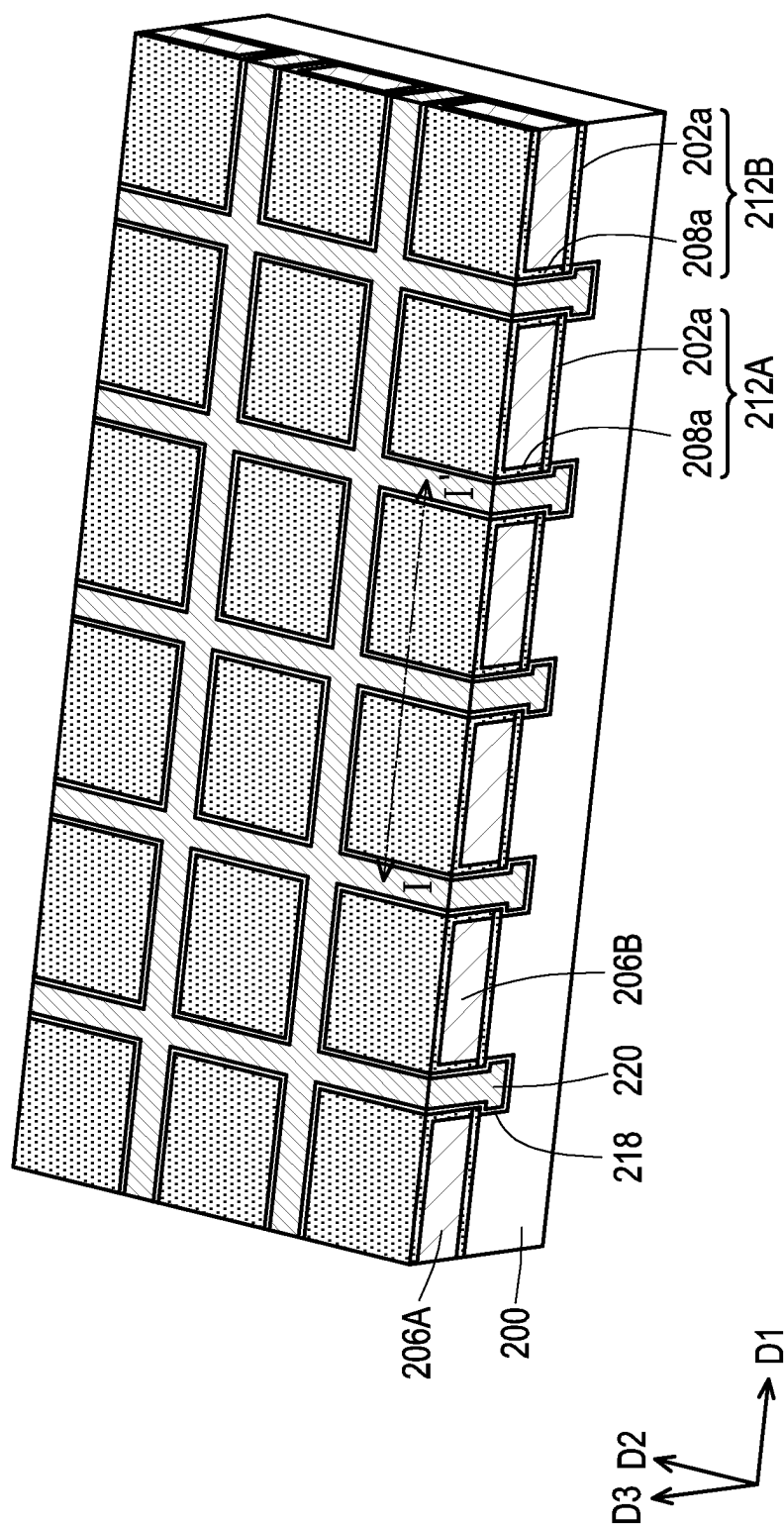


FIG. 2G

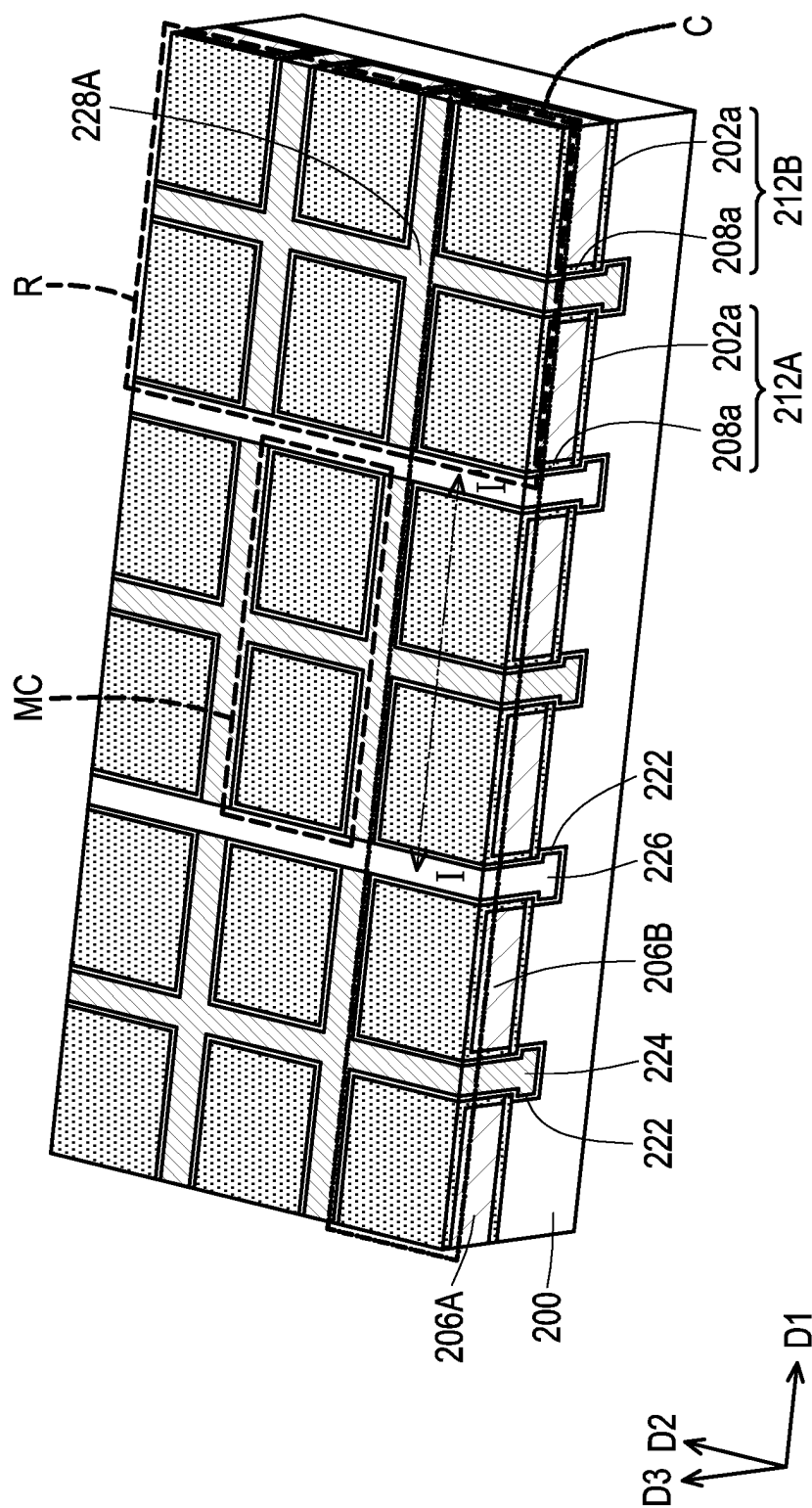


FIG. 2H

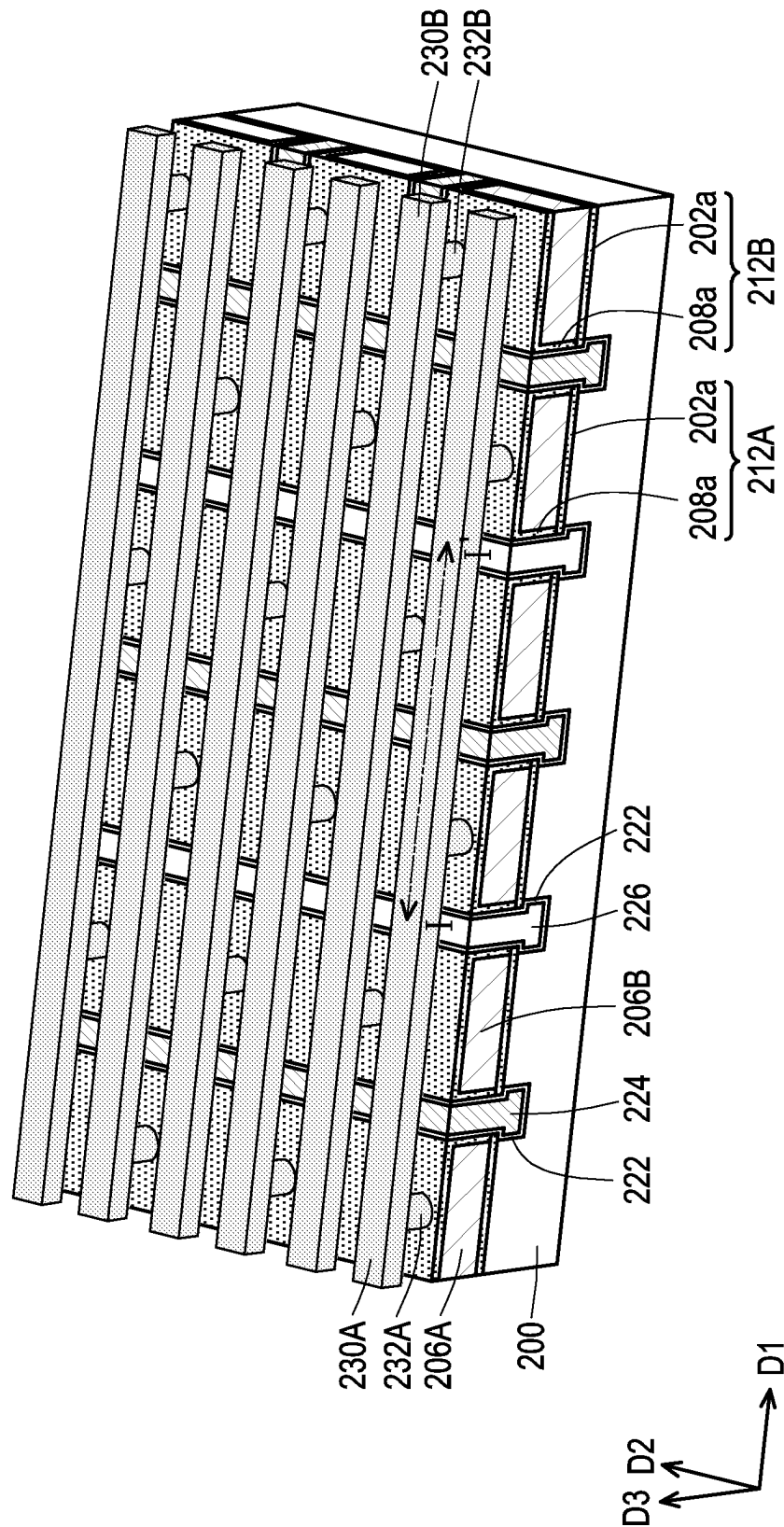


FIG. 21

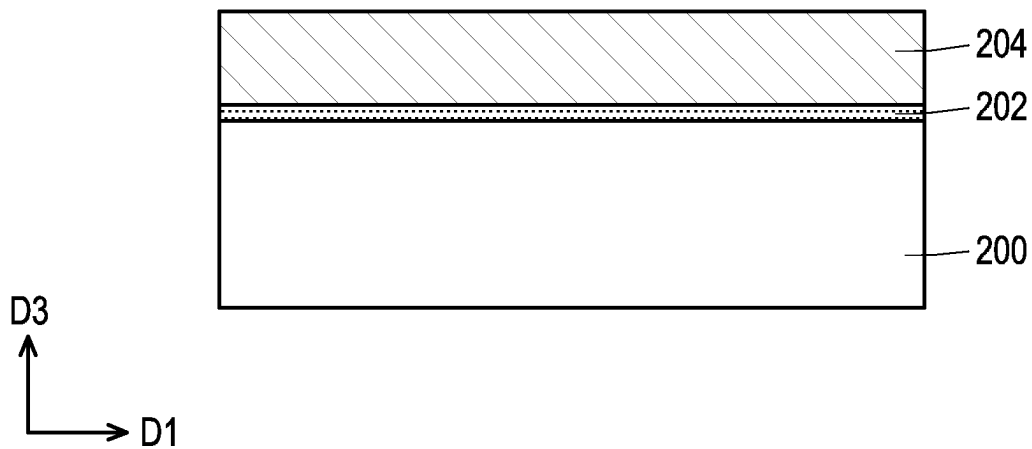


FIG. 3A

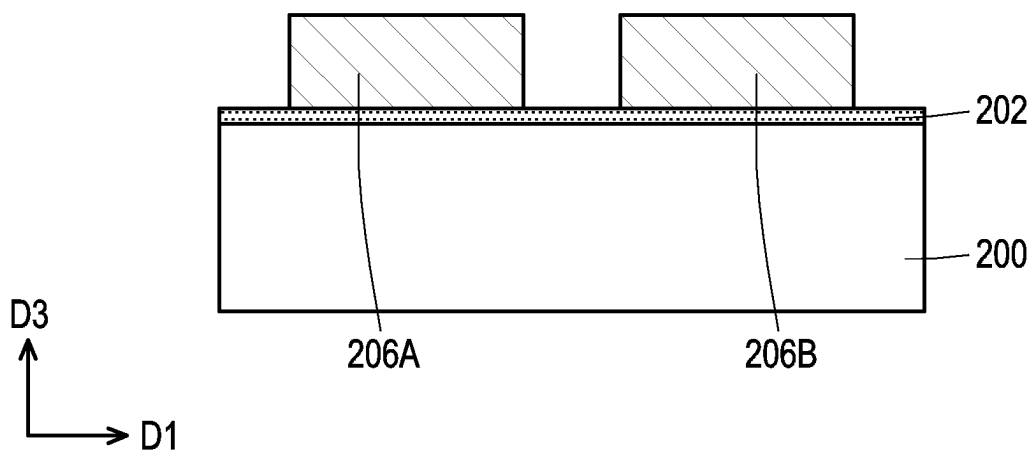


FIG. 3B

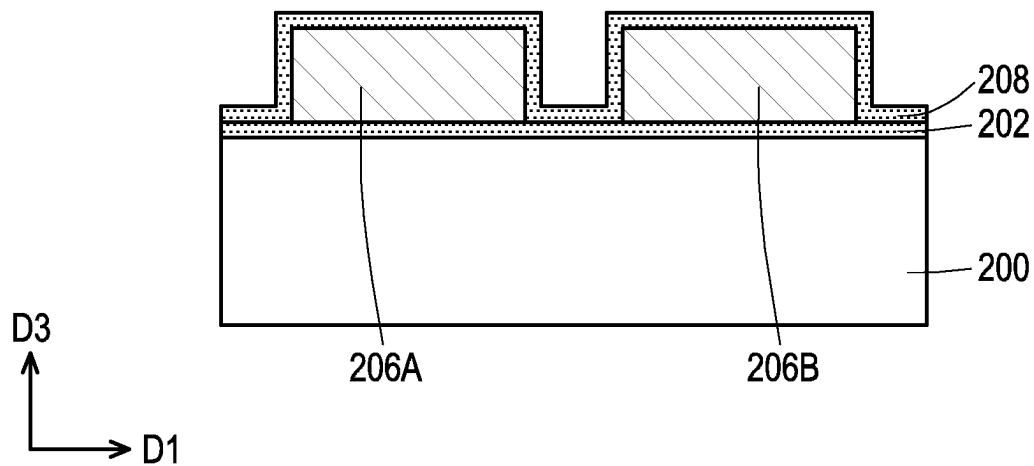


FIG. 3C

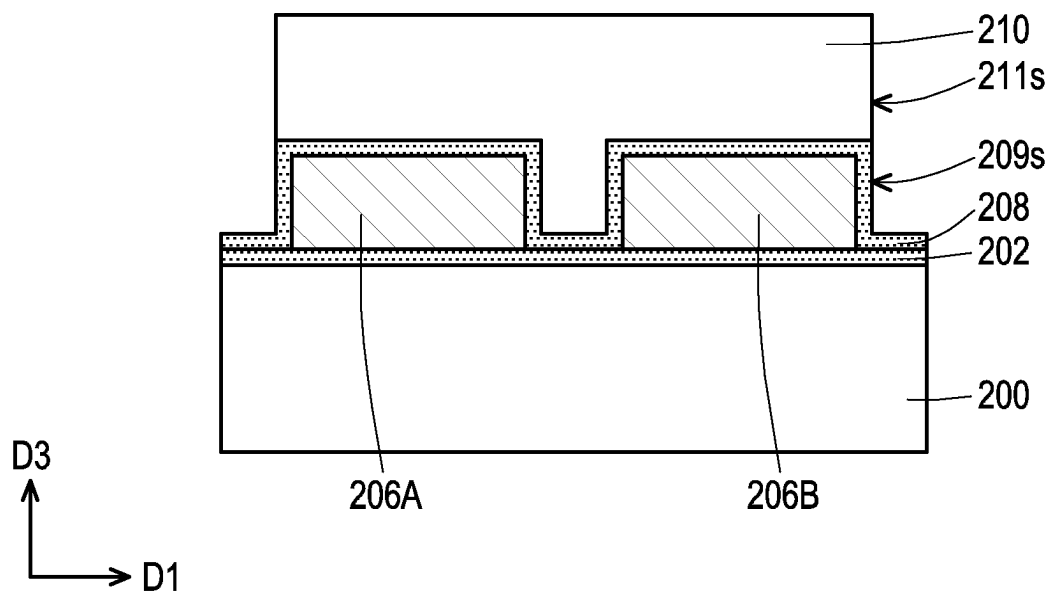


FIG. 3D

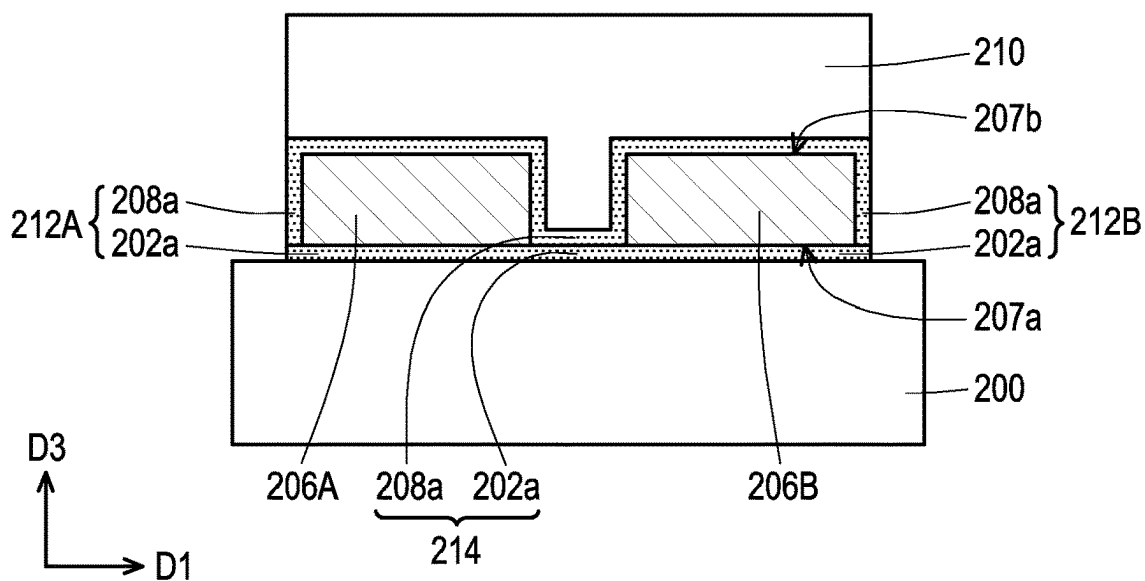


FIG. 3E

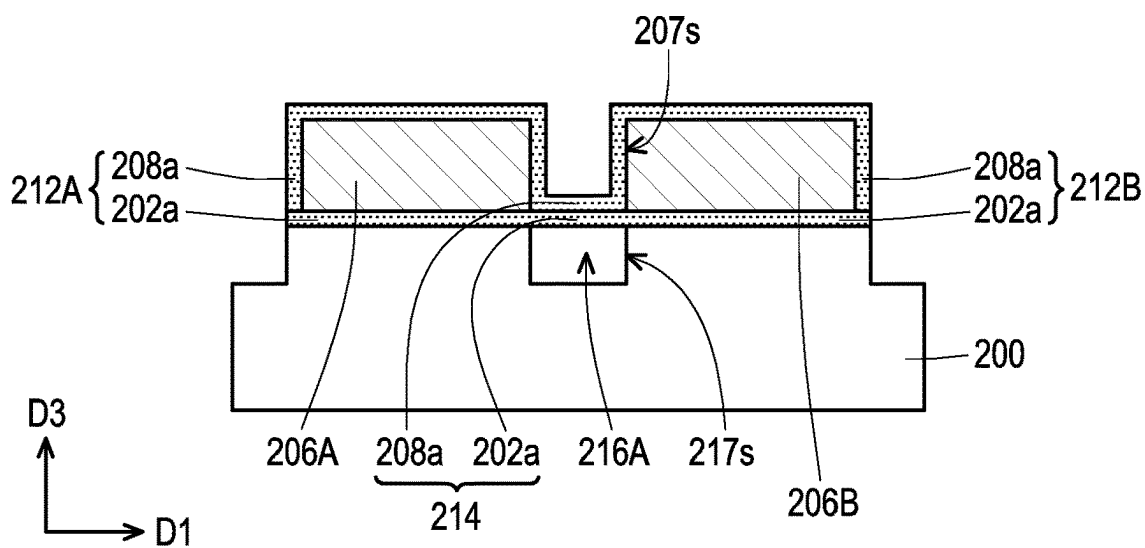


FIG. 3F

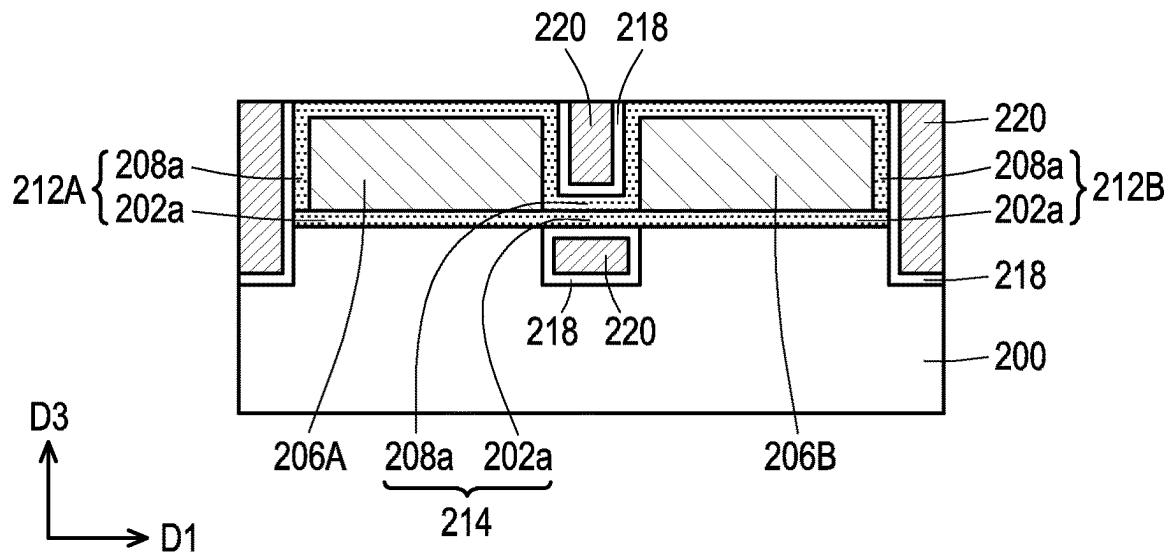


FIG. 3G

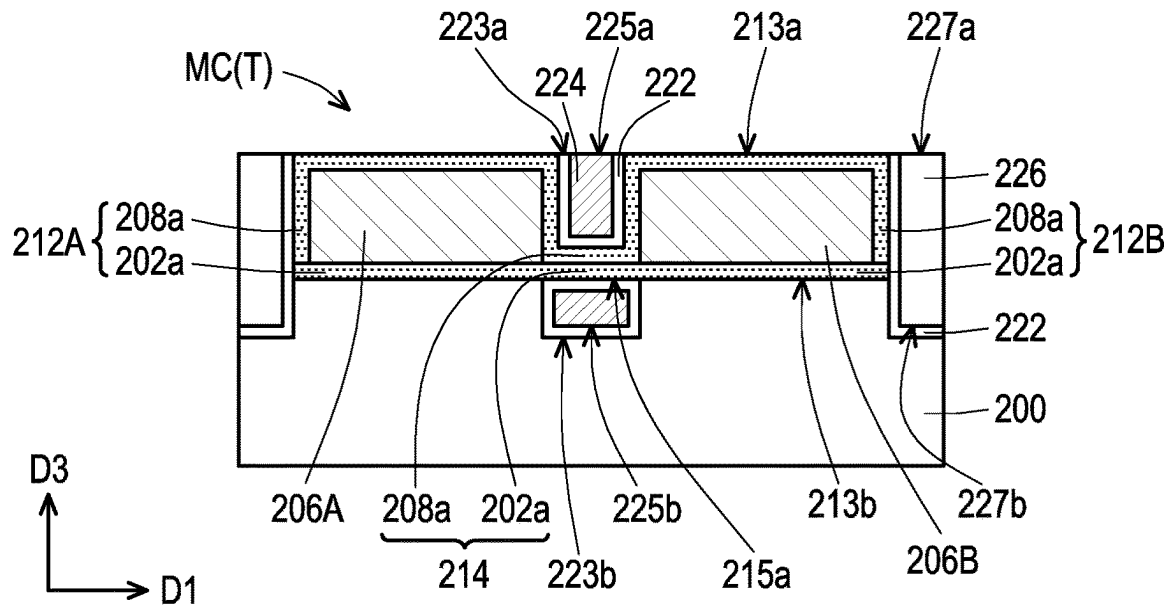


FIG. 3H

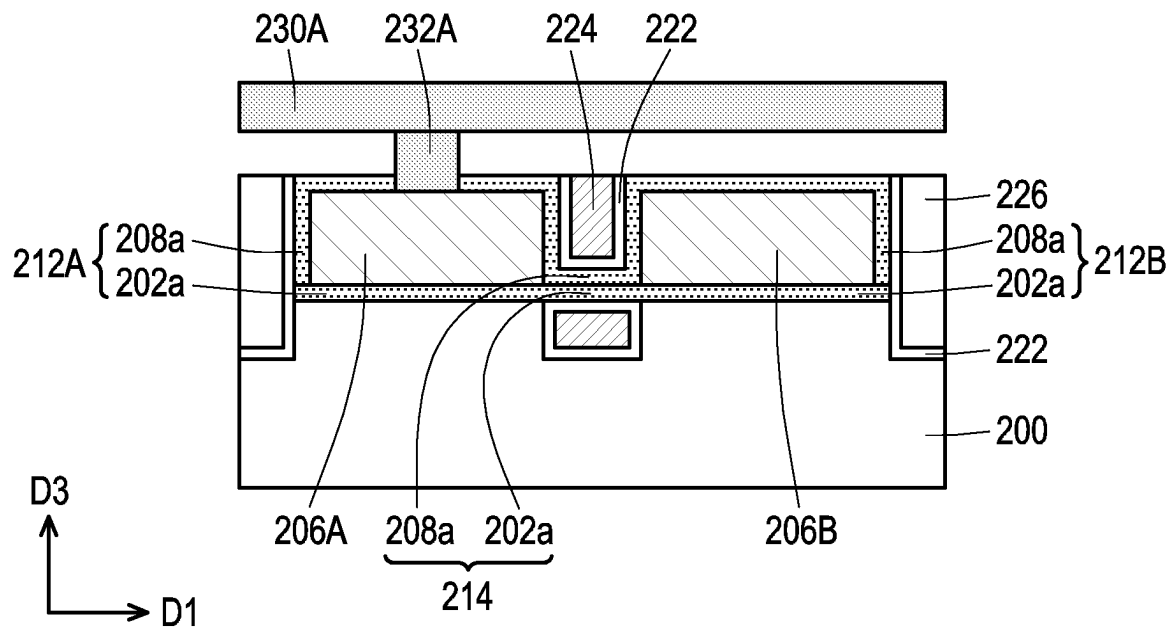


FIG. 3I

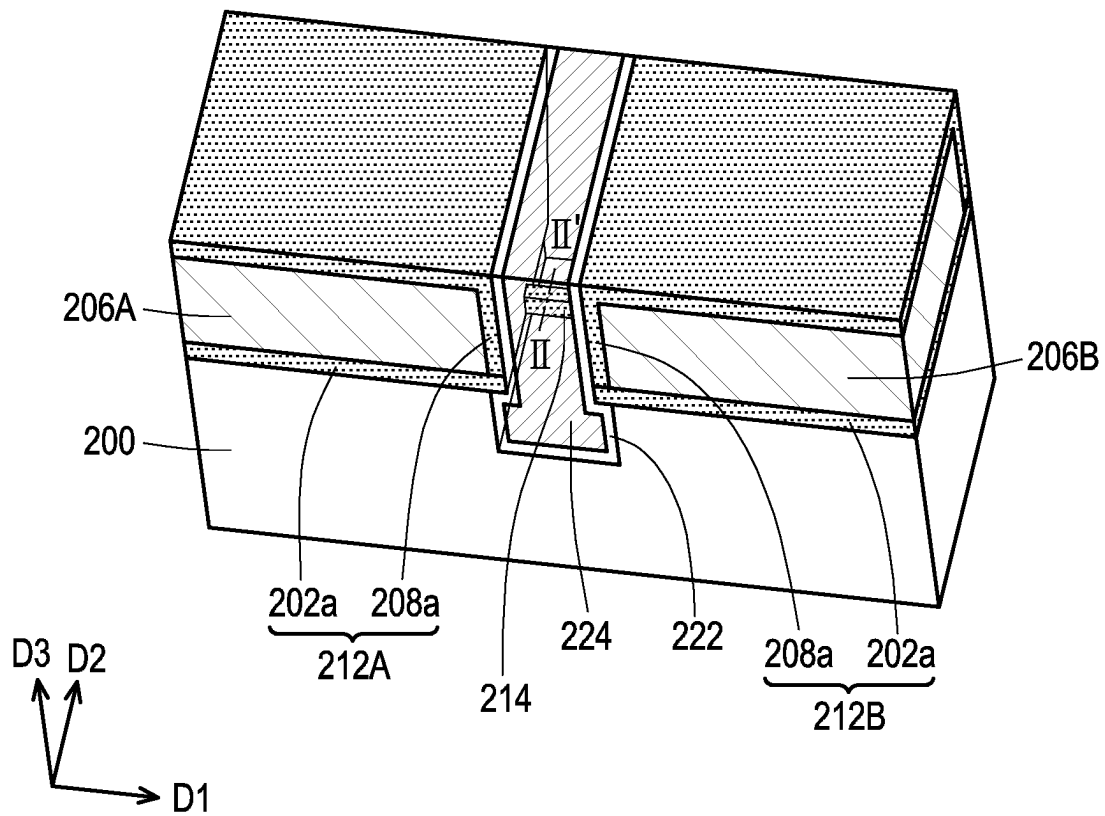


FIG. 4A

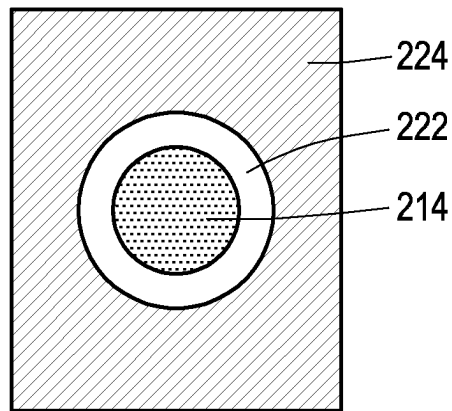


FIG. 4B

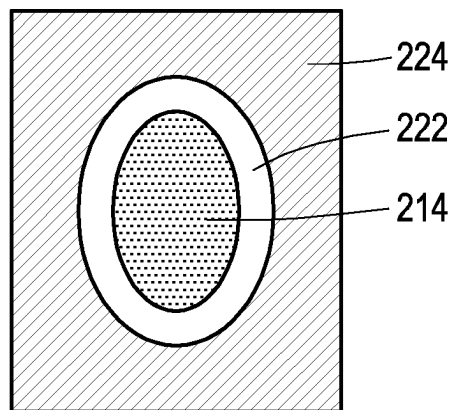


FIG. 5A

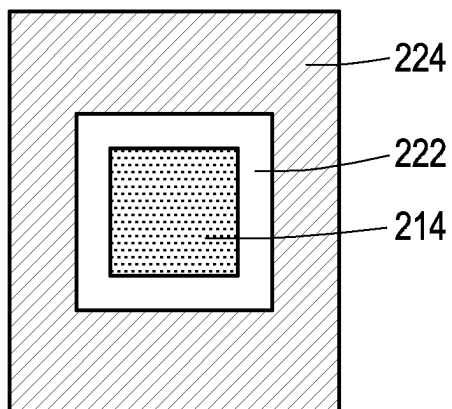


FIG. 5B

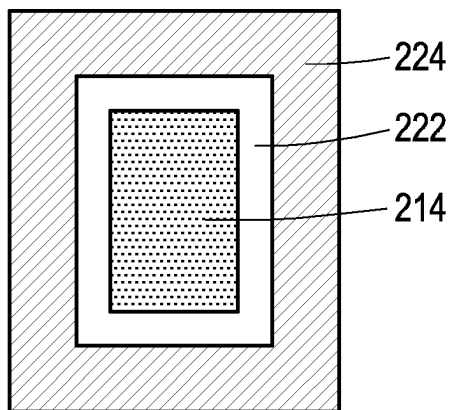


FIG. 5C

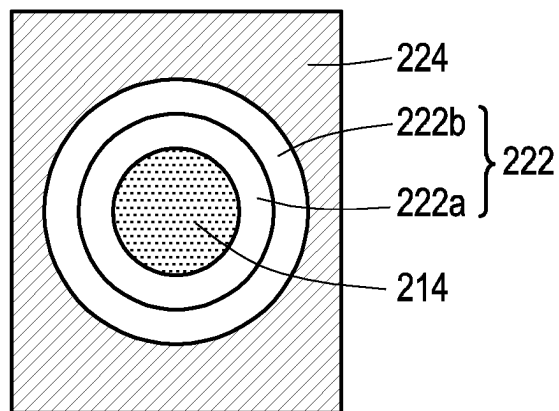


FIG. 5D

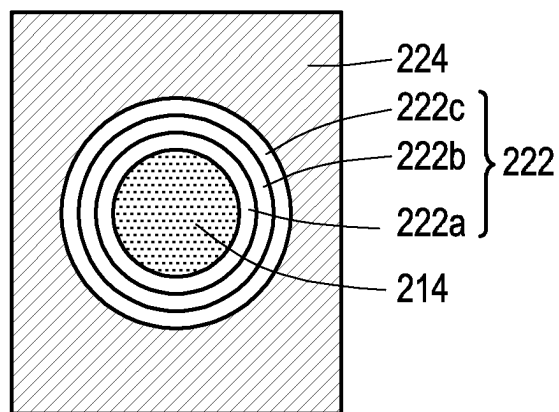


FIG. 5E

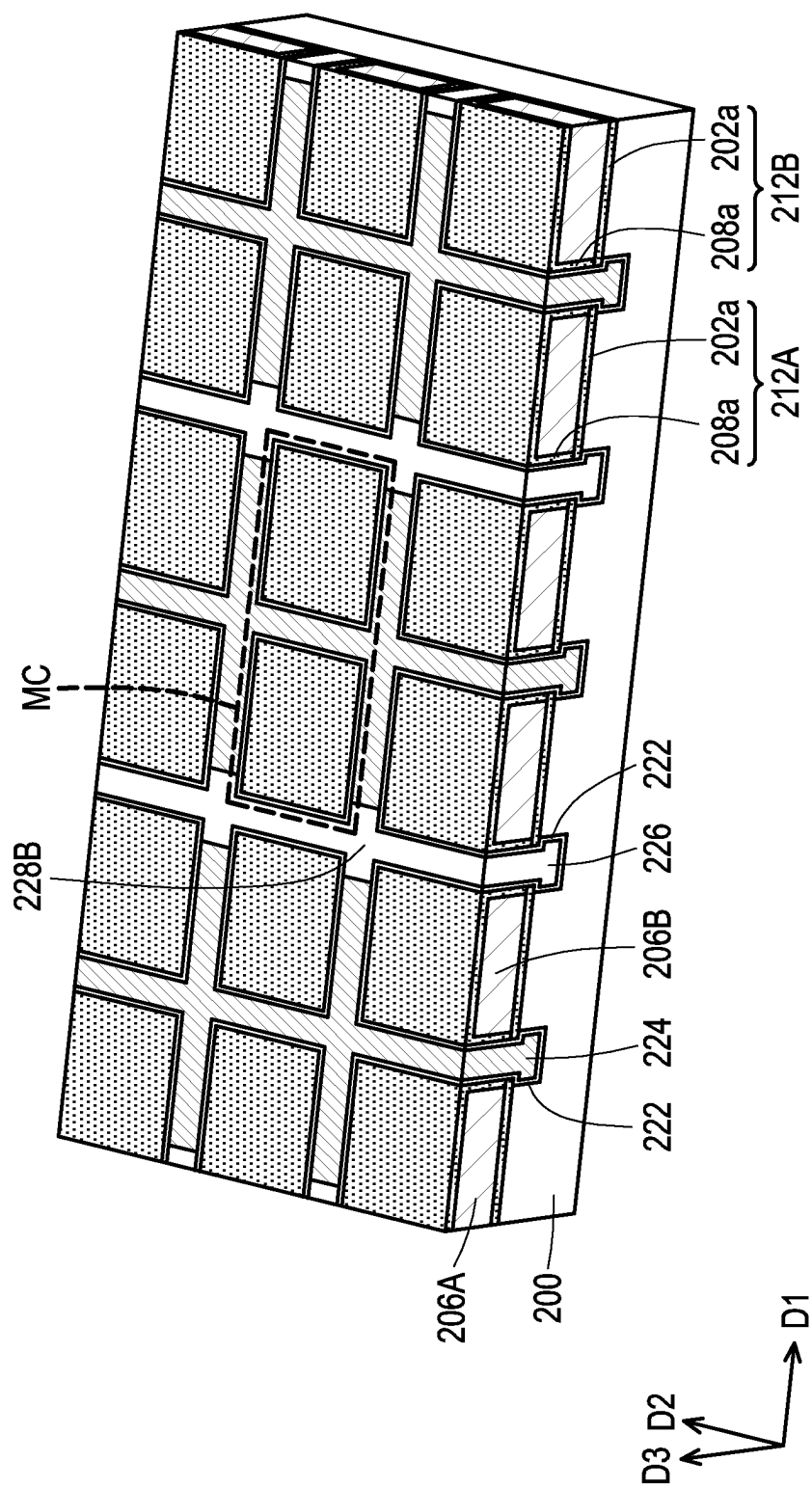


FIG. 6

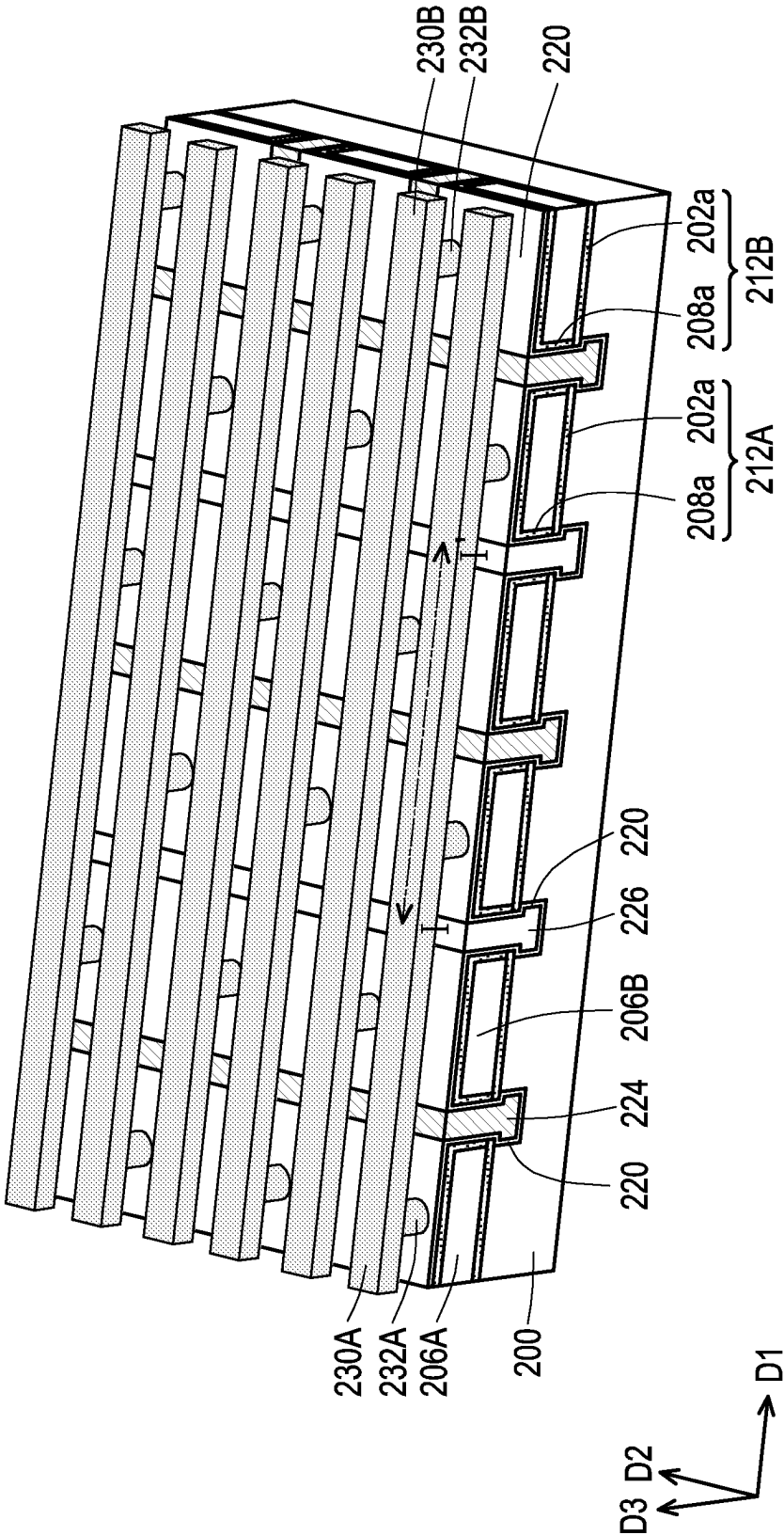


FIG. 7A

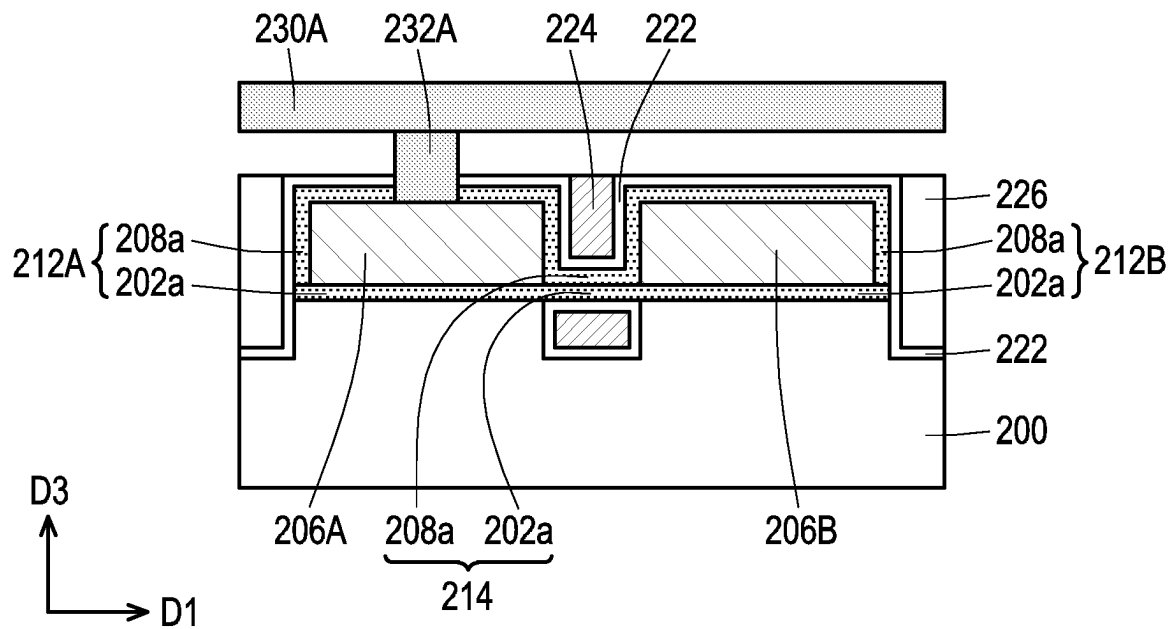


FIG. 7B

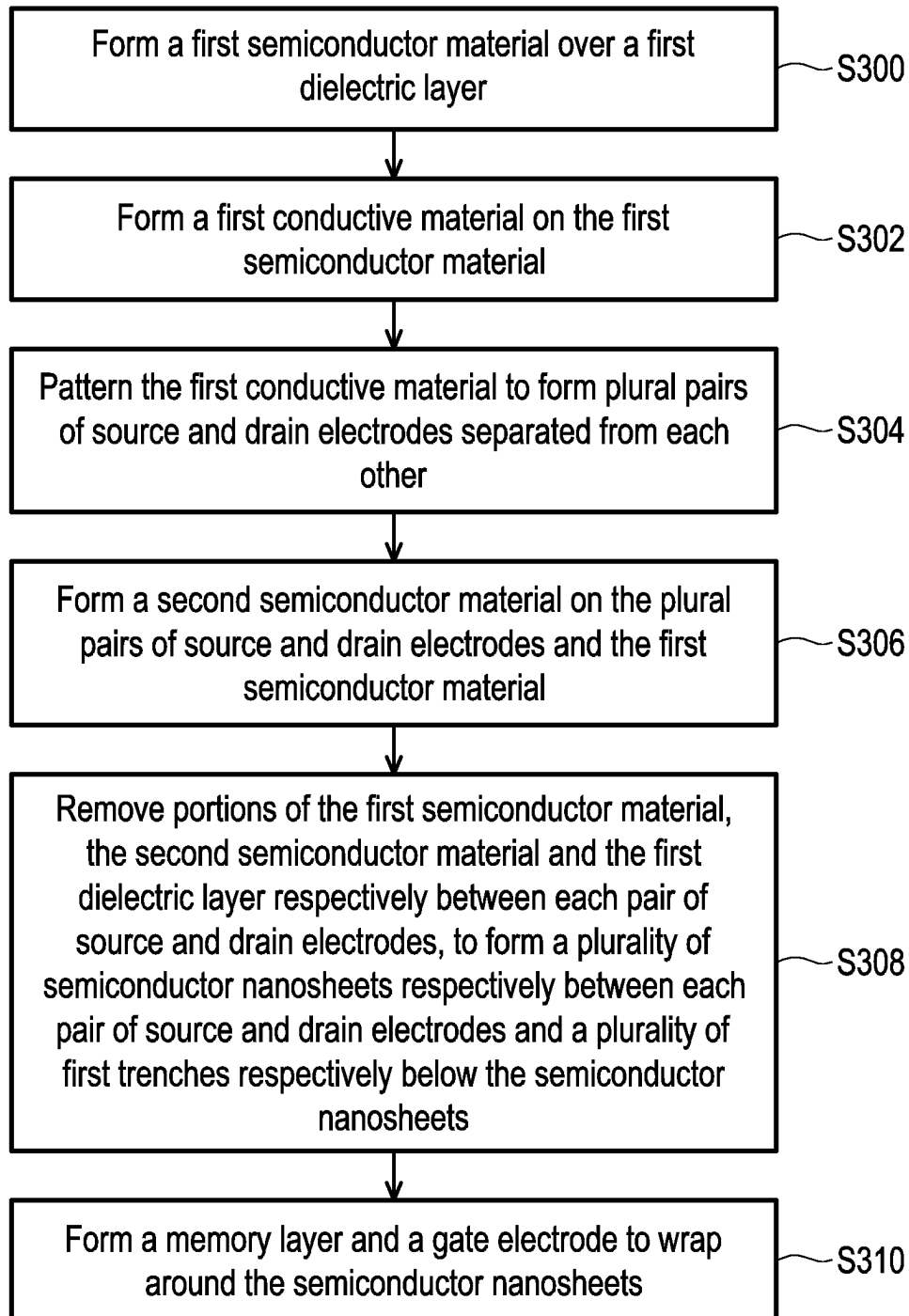


FIG. 8

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TRANSISTOR, MEMORY DEVICE AND MANUFACTURING METHOD OF MEMORY DEVICE

BACKGROUND

As the size of the integrated circuit keeps decreasing, the integration density of the component or device gradually increases. Semiconductor memory devices include volatile memories and non-volatile memories. For semiconductor memory devices, the increased memory cell density leads to compact structure designs with reduced sizes but maintaining the performance of the semiconductor memory devices.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 illustrates a cross-sectional view of a semiconductor device in accordance with some embodiments.

FIG. 2A to FIG. 2I are three dimensional views illustrating various stages of a manufacturing method of a memory device.

FIG. 3A to FIG. 3I are cross-sectional views taken along cross section line I-I' in FIG. 2A to FIG. 2I.

FIG. 4A is a perspective three dimensional view of a portion of FIG. 2H, and FIG. 4B is a cross-sectional view taken along a cross section line II-II' in FIG. 4A.

FIG. 5A to FIG. 5E are cross-sectional views taken along a cross section line II-II' in FIG. 4A.

FIG. 6 is a cross-sectional view of a memory device in accordance with some alternative embodiments of the disclosure.

FIG. 7A is a three dimensional view of a memory device, and FIG. 7B is a cross-sectional view taken along cross section line I-I' in FIG. 7A.

FIG. 8 illustrates a manufacturing method of a memory device in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s)

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as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

The gate all around (GAA) transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

FIG. 1 illustrates a cross-sectional view of a semiconductor device 10 in accordance with some embodiments. In some embodiments, the semiconductor device 10 is formed with integrated memory devices 120 and 130. The semiconductor device 10 may include active devices 110 and memory devices 120, 130. The active devices 110 may be field effect transistor (FET) devices. In one embodiment, the active devices 110 are formed through the front-end-of-line (FEOL) manufacturing processes and include fin field effect transistors (FinFETs). The at least one of the memory devices 120, 130 may include ferroelectric random access memory (FeRAM) devices formed through the back-end-of-line (BEOL) manufacturing processes. It is understood that FinFETs are used as examples, and other kinds of FEOL devices such as planar transistors or gate-all-around (GAA) transistors may be used herein and included within the scope of the present disclosure. That is, the memory devices 120, 130 may be integrated with or in any suitable semiconductor devices. In FIG. 1, the details of the memory devices 120, 130 are not shown and further details will be described later in subsequent figures.

As illustrated in FIG. 1, the semiconductor device 10 includes different regions for forming different types of circuits. For example, the semiconductor device 10 includes a first region 102 for forming logic circuits and a second region 104 for forming peripheral circuits, input/output (I/O) circuits, electrostatic discharge (ESD) circuits, and/or analog circuits. The semiconductor device 10 may also include other regions for forming other types of circuits which are fully intended to be included within the scope of the present disclosure. The semiconductor device 10 includes a substrate 101. In some embodiments, the substrate 101 is a bulk substrate, such as a silicon substrate, doped or undoped, or an active layer of a semiconductor-on-insulator (SOI) substrate. In some embodiments, the substrate 101 includes other semiconductor materials, such as germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, gallium nitride, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. Other substrates, such as multi-layered or gradient substrates, may also be used. For example, additional electrical components, such as resistors, capacitors, inductors, diodes,

or the like, is formed in or on the substrate **101** during the FEOL manufacturing processes.

As seen in FIG. 1, the active devices **110** are formed on the substrate **101**, and isolation regions **103**, such as shallow trench isolation (STI) regions, are formed between or around the active devices **110**. In some embodiments, the active device **110** includes a gate electrode **107** and source and drain regions **105** and **106**. The gate electrode **107** may be formed over the substrate **101** with gate spacers **108** along sidewalls of the gate electrode **107**. The source and drain regions **105** and **106** such as doped or epitaxial source and drain regions are formed on opposing sides of the gate electrode **107**. In some embodiments, conductive contacts **109**, such as gate contacts and source/drain contacts, are formed over and electrically coupled to respective underlying electrically conductive features (e.g., gate electrodes **107** or source and drain regions **105** and **106**). In some embodiments, a dielectric layer **116**, such as an inter-layer dielectric (ILD) layer, is formed over the substrate **101** and covering the source and drain regions **105** and **106**, the gate electrode **107** and the contacts **109**, and other electrically conductive features, such as conductive interconnect structures including conductive vias **112** and conductive lines **114**, are embedded in the dielectric layer **116**. It is understood that the dielectric layer **116** may include more than one dielectric layers of the same or different dielectric materials. Collectively, the substrate **101**, the active devices **110**, the contacts **109**, conductive features **112/114**, and the dielectric layers **116** shown in FIG. 1 may be referred to as the front-end level **12L**.

Referring to FIG. 1, dielectric layers **118** and dielectric layers **122** are formed over the dielectric layer **116** in alternation. In one embodiment, at least one of the dielectric layers **118** includes an etch stop layer (ESL). In some embodiments, the materials of the dielectric layers **118** is different from the materials of the dielectric layers **116** and **122**. In some embodiments, the material of the dielectric layer(s) **118** includes silicon nitride or carbide formed by plasma-enhanced physical vapor deposition (PECVD). In some embodiments, one or more of the dielectric layers **118** is omitted. In some embodiments, the dielectric layers **116** and **122** is formed of any suitable dielectric material, such as silicon oxide, silicon nitride, silicon oxynitride, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), or low-k materials, formed by a suitable method, such as spin coating, physical vapor deposition (PVD), chemical vapor deposition (CVD), or the like. In FIG. 1, memory devices **120** and **130**, each of which may include a plurality of memory cells, are formed in the dielectric layer **122** and coupled to electrically conductive features (e.g., conductive vias **124** and conductive lines **125**) in the dielectric layer **122**.

In FIG. 1, the memory devices **120** and **130** are formed at different layers of the dielectric layers **122**. The memory device **120** is formed at the lower layer, and the memory device **130** is formed at the upper layer. In some embodiments, the memory devices **120** and **130** have the same or similar structure. In some embodiments, the memory devices **120** and **130** have different structure designs. Although two layers of memory devices are depicted in FIG. 1, other numbers of layers of memory devices, such as one layer, three layers, or more, are also possible and are encompassed within the scope of the present disclosure. Collectively, the layers of memory device **120** and **130** are referred to as the memory device level **14L** or a memory region of the semiconductor device **10**. The memory device level **14L** may be formed in the BEOL processes of semi-

conductor manufacturing. The memory devices **120** and **130** may be formed in the BEOL processes at any suitable locations within the semiconductor device **10**, such as over the first region **102**, over the second region **104**, or over a plurality of regions.

After the memory device level **14L** is formed, an interconnect level **16L** including electrically conductive interconnecting features (e.g., conductive vias **126** and conductive lines **127**) embedded in the dielectric layer(s) **122** is formed over the memory device level **14L**. Any suitable methods may be employed to form the interconnect level **16L**, and the details are not described herein. In some embodiments, the interconnect level **16L** is electrically connect the electrical components formed in/on the substrate **101** to form functional circuits. In some embodiments, the interconnect structure **16L** is also electrically coupled the memory devices **120**, **130** to the active devices **110** and/or the components in/on the substrate **101**. In addition, the memory devices **120** and **130** may be electrically coupled to an external circuit or an external device through the structure of the interconnect level **16L**. In some embodiments, the memory devices **120** and **130** are electrically coupled to the active devices **110** of the front-end level **12L** and/or other electrical components formed in the substrate **101**, and are controlled or accessed (e.g., written to or read from) by functional circuits of the semiconductor device **10**. Alternatively, the memory devices **120**, **130** are electrically coupled to (e.g., controlled or accessed) an external circuit of another semiconductor device through the structure of the interconnect level **16L**.

FIG. 2A to FIG. 2I are three dimensional views illustrating various stages of a manufacturing method of a memory device. FIG. 3A to FIG. 3I are cross-sectional views taken along cross section line I-I' in FIG. 2A to FIG. 2I. FIG. 4A is a perspective three dimensional view of a portion of FIG. 2H, and FIG. 4B is a cross-sectional view taken along cross section line II-II' in FIG. 4A. According to some embodiments, the memory device may be a three-dimensional (3D) memory device with a ferroelectric material. The memory devices depicted in the following paragraphs may be used as the memory devices **120** and **130** in FIG. 1.

Referring to FIG. 2A and FIG. 3A, a semiconductor material **202** is formed on a dielectric layer **200**. In some embodiments, the dielectric layer **200** is the dielectric layer(s) **122** in the memory device level **14L** of FIG. 1, so the detailed description thereof is omitted herein. The dielectric layer **200** is formed over the front-end level **12L**. The dielectric layer **200** may be a single layer or multiple layers. In some embodiments, the dielectric layer **200** is also referred to as a buffer layer. A material of the dielectric layer **200** is not particularly limited, as long as said material renders good etching selectivity between the semiconductor material **202** and the dielectric layer **200** and between a semiconductor material **208** (described below) and the dielectric layer **200**. For example, the dielectric layer **200** is made of polyimide, epoxy resin, acrylic resin, phenol resin, BCB, PBO, silicon oxide, silicon nitride, or any other suitable polymer-based dielectric material.

In some embodiments, the semiconductor material **202** is made of a first conductive type oxide semiconductor material. The first conductive type is n-type or p-type. In some embodiments, the semiconductor material **202** includes n-type amorphous oxide semiconductor such as amorphous indium tungsten oxide (a-IWO), amorphous indium zinc oxide (a-IZO), amorphous indium-tungsten-zinc oxide (a-IWZO), amorphous indium-tin-zinc oxide (a-ITZO), amorphous indium tin oxide (a-ITO), amorphous indium

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oxide (a-InO) material, the like, or a combination thereof. In alternative embodiments, the semiconductor material **202** includes p-type oxide based material such as SnO_x , Cu_xO (e.g., Cu_2O , CuFeO_2 , $\text{Cu}_{15}\text{Fe}_3\text{O}_2$), NiO_x , $\text{Ni}(\text{Sn})\text{O}$, the like, or a combination thereof. In some embodiments, the semiconductor material **202** is doped with a first conductive type dopant. For example, the semiconductor material **202** is doped with p-type dopants, such as boron, BF_2 , the like, or a combination thereof, or doped with n-type dopants, such as phosphorus, arsenic, the like, or a combination thereof. In some embodiments, the semiconductor material **202** includes IV element such as Ge, Si—Ge, Ge—Si, SiC or Ge—Sn, a compound such as GaN, GaAs, GaP, GaSb, InN, InAs, InSb, BN, BP, AlN, AlP, AlAs, AlSb, CdSe, CdS, CdTe, ZnS or ZnTe, 2D material such as graphene, MoS_2 , MoTe, MoSe, WSe₂, WS₂, h-BN or PbI_2 , the like, or a combination thereof. In some embodiments, an etching selectivity between the semiconductor material **202** and the dielectric layer **200** is high. For example, the etching selectivity between the semiconductor material **202** and the dielectric layer **200** ranges between 1:10 and 1:10000. Herein, the etching selectivity is denoted by a ratio between an etch rate of the semiconductor material **202** and the dielectric layer **200**. In some embodiments, the semiconductor material **202** is made of a single layer having one of the foregoing materials. However, the disclosure is not limited thereto. In some alternative embodiments, the semiconductor material **202** is made of a laminate structure of at least two of the foregoing materials. In some embodiments, the semiconductor material **202** is deposited on the dielectric layer **200** through ALD, CVD, PVD, or the like. In some embodiments, a thickness of the semiconductor material **202** is in a range of 0.5 nm to 10 nm.

Then, a conductive material **204** is formed on the semiconductor material **202** over the dielectric layer **200**. In some embodiments, the conductive material **204** includes cobalt, tungsten, copper, titanium, tantalum, aluminum, zirconium, hafnium, the like, a combination thereof, or other suitable conductive materials. In some embodiments, the conductive material **204** is formed through CVD, ALD, plating, or other suitable deposition techniques. In some embodiments, a barrier layer (not shown) is optionally formed between the conductive material **204** and the semiconductor material **202**, so as to avoid diffusion of atoms between elements. The barrier layer includes, for example, TiN, TaN, TiSiN, TaSiN, WSiN, TiC, TaC, TiAlC, TaAlC, TiAlN, TaAlN, or a combination thereof.

Referring to FIG. 2B and FIG. 3B, the conductive material **204** is patterned, to form a plurality of source and drain electrodes **206A** and **206B**. For example, portions of conductive material **204** are removed to expose the underlying semiconductor material **202**. In some embodiments, the conductive material **204** is patterned through a lithography process and an etching process by using a mask. The lithography process includes, for example, photoresist coating, soft baking, exposing, post-exposure baking (PEB), developing, and hard baking. The etching process includes, for example, an anisotropic etching process such as dry etch or an isotropic etching process such as wet etch.

One of the source and drain electrodes **206A** and **206B** is a source electrode (e.g., source electrode **206A**), and the other of the source and drain electrodes **206A** and **206B** is a drain electrode (e.g., drain electrode **206B**), and vice versa. In some embodiments, the source and drain electrodes **206A** and **206B** are physically separated from each other. In some embodiments, the adjacent source and drain electrodes **206A** and **206B** arranged along a direction D1 are paired, and thus

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are also referred to as a pair of source and drain electrodes **206A** and **206B**. In some embodiments, plural pairs of source and drain electrodes **206A** and **206B** are arranged in an array having a plurality of rows R arranged along the direction D1 and a plurality of columns C arranged along a direction D2 substantially perpendicular to the direction D1. For example, each row R of the array has plural pairs of source and drain electrodes **206A** and **206B** arranged along the direction D2, and each column of the array has plural pairs of source and drain electrodes **206A** and **206B** arranged along the direction D1. The direction D1 and the direction D2 are, for example, substantially perpendicular to a stacking direction D3 of the dielectric layer **200**, the semiconductor material **202** and the source and drain electrodes **206A** and **206B**. For example, the direction D1 is x-direction, the direction D2 is y-direction, and the direction D3 is z-direction. The source and drain electrodes **206A** and **206B** cover portions of the semiconductor material **202** while portions of the semiconductor material **202** between the source and drain electrodes **206A** and **206B** are exposed. In some embodiments, the source electrodes **206A** are also referred to as source lines, and the drain electrodes **206B** are also referred to as bit lines.

Referring to FIG. 2C and FIG. 3C, a semiconductor material **208** is formed over the semiconductor material **202**, to cover the source and drain electrodes **206A** and **206B** and the exposed semiconductor material **202**. In some embodiments, the semiconductor material **208** has the same material of the semiconductor material **202**. In alternative embodiments, the semiconductor material **208** has different material from the semiconductor material **202** as long as they have the same conductive type. In some embodiments, the semiconductor material **208** is made of a first conductive type oxide semiconductor material. The first conductive type is n-type or p-type. In some embodiments, the semiconductor material **208** includes n-type amorphous oxide semiconductor such as amorphous indium tungsten oxide (a-IWO), amorphous indium zinc oxide (a-IZO), amorphous indium-tungsten-zinc oxide (a-IWZO), amorphous indium-tin-zinc oxide (a-ITZO), amorphous indium tin oxide (a-ITO), amorphous indium oxide (a-InO) material, the like, or a combination thereof. In alternative embodiments, the semiconductor material **208** includes p-type oxide based material such as SnO_x , Cu_xO (e.g., Cu_2O , CuFeO_2 , $\text{Cu}_{15}\text{Fe}_3\text{O}_2$), NiO_x , $\text{Ni}(\text{Sn})\text{O}$, the like, or a combination thereof. In some embodiments, the semiconductor material **208** is doped with a first conductive type dopant. For example, the semiconductor material **208** is doped with p-type dopants, such as boron, BF_2 , the like, or a combination thereof, or doped with n-type dopants, such as phosphorus, arsenic, the like, or a combination thereof. In some embodiments, the semiconductor material **208** includes IV element such as Ge, Si—Ge, Ge—Si, SiC or Ge—Sn, a compound such as GaN, GaAs, GaP, GaSb, InN, InAs, InSb, BN, BP, AlN, AlP, AlAs, AlSb, CdSe, CdS, CdTe, ZnS or ZnTe, 2D material such as graphene, MoS_2 , MoTe, MoSe, WSe₂, WS₂, h-BN or PbI_2 , the like, or a combination thereof. In some embodiments, an etching selectivity between the semiconductor material **208** and the dielectric layer **200** is high. For example, the etching selectivity between the semiconductor material **208** and the dielectric layer **200** ranges between 1:10 and 1:10000. Herein, the etching selectivity is denoted by a ratio between an etch rate of the semiconductor material **202** and the dielectric layer **200**. In some embodiments, the semiconductor material **208** is made of a single layer having one of the foregoing materials. However, the disclosure is not limited thereto. In some alternative embodiments, the semiconduc-

tor material **208** is made of a laminate structure of at least two of the foregoing materials. In some embodiments, the semiconductor material **208** is deposited on the dielectric layer **200** through ALD, CVD, PVD, or the like. In some embodiments, a thickness of the semiconductor material **208** is in a range of 0.5 nm to 10 nm.

In some embodiments, the semiconductor material **208** is conformally formed on the source and drain electrodes **206A** and **206B** and the exposed semiconductor material **202**. For example, the semiconductor material **208** is continuously disposed on and in direct contact with all exposed surfaces of the source and drain electrodes **206A** and **206B** and the semiconductor material **202**. In some embodiments, the semiconductor material **208** surrounds the source and drain electrodes **206A** and **206B**.

Referring to FIG. 2D, FIG. 2E, FIG. 3D and FIG. 3E, portions of the semiconductor materials **202** and **208** between the source and drain electrodes **206A** and **206B** are removed. In some embodiments, the semiconductor materials **202** and **208** are partially removed through an etching process by using a patterned photoresist layer **210**. In some embodiments, as shown in FIG. 2D and FIG. 3D, the patterned photoresist layer **210** is formed on the semiconductor material **208**, to cover the source and drain electrodes **206A** and **206B** and regions respectively between a pair of source and drain electrodes **206A** and **206B**. As shown in FIG. 2D, the patterned photoresist layer **210** may cover plural pairs of source and drain electrodes **206A** and **206B** in the same row R, and expose the semiconductor material **208** between adjacent rows R. For example, as shown in FIG. 3D, sidewalls **211s** of the patterned photoresist layer **210** are substantially flush with sidewalls **209s** of the semiconductor material **208**.

Then, as shown in FIG. 2E and FIG. 3E, by using the patterned photoresist layer **210** as a mask, portions of the semiconductor materials **202** and **208** are removed, to form a semiconductor layer **212A**, a semiconductor layers **212B** and a semiconductor nanosheet **214**. In some embodiments, the semiconductor layer **212A** surrounds one of a pair of source and drain electrodes **206A** and **206B**, the semiconductor layers **212B** surrounds the other of a pair of source and drain electrodes **206A** and **206B** and the semiconductor nanosheet **214** is disposed between a pair of source and drain electrodes **206A** and **206B**. For example, the semiconductor layer **212A** surrounds the source electrode **206A** while the semiconductor layer **212B** surrounds the drain electrode, and vice versa. In some embodiments, the semiconductor materials **202** and **208** are removed through an etching process. The etching process includes, for example, an isotropic etching process such as wet etch or an anisotropic etching process such as dry etch. In some embodiments, an etchant for the wet etch includes a combination of hydrogen fluoride (HF) and ammonia (NH₃), a combination of HF and tetramethylammonium hydroxide (TMAH), or the like. On the other hand, the dry etch process includes, for example, reactive ion etch (RIE), inductively coupled plasma (ICP) etch, electron cyclotron resonance (ECR) etch, neutral beam etch (NBE), and/or the like.

In some embodiments, the semiconductor layer **212A**, the semiconductor layer **212B** and the semiconductor nanosheet **214** are respectively include a semiconductor layer **202a** and a semiconductor layer **208a**. For example, the semiconductor layer **202a** of the semiconductor layer **212A** is disposed at a first surface (e.g., bottom surface) **207a** of one (e.g., the source electrode **206A**) of a pair of source and drain electrodes **206A** and **206B**, and the semiconductor layer **208a** of the semiconductor layer **212A** is disposed at a second

surface (e.g., top surface) **207b** opposite to the first surface of the one (e.g., source electrode **206A**) of the pair of source and drain electrodes **206A** and **206B** and on sidewalls between the first and second surfaces **207a** and **207b**. Similarly, the semiconductor layer **202a** of the semiconductor layer **212B** is disposed at a first surface (e.g., bottom surface) **207a** of the other (e.g., drain electrode **206B**) of the pair of source and drain electrodes **206A** and **206B**, and the semiconductor layer **208a** of the semiconductor layer **212B** is disposed at a second surface (e.g., top surface) **207b** opposite to the first surface of the other (e.g., drain electrode **206B**) of the pair of source and drain electrodes **206A** and **206B** and on sidewalls between the first and second surfaces **207a** and **207b**. In other words, one of a pair of source and drain electrodes **206A** and **206B** is surrounded by the semiconductor layer **212A**, and the other of the pair of source and drain electrodes **206A** and **206B** is surrounded by the semiconductor layer **212B**. The semiconductor nanosheet **214** includes the semiconductor layer **202a** and the semiconductor layer **208a** on the semiconductor layer **202a**, for example.

In some embodiments, the semiconductor layer **202a** is continuously formed on a pair of source and drain electrodes **206A** and **206B** and the region between the pair of source and drain electrodes **206A** and **206B**, and the semiconductor layer **208a** is also continuously formed below a pair of source and drain electrodes **206A** and **206B** and the region between the pair of source and drain electrodes **206A** and **206B**. In some embodiments, the semiconductor layer **212A** and the semiconductor layer **212B** are physically connected by the semiconductor nanosheet **214** therebetween. In some embodiments, the semiconductor layer **212A**, the semiconductor nanosheet **214** and the semiconductor layer **212B** are continuous and thus may be referred to as a semiconductor structure (or a channel structure), and the source and drain electrodes **206A** and **206B** are embedded in the semiconductor structure. In an embodiment in which the semiconductor layer **202a** and the semiconductor layer **208a** have the same material, an interface does not exist between the semiconductor layer **202a** and the semiconductor layer **208a**. For example, the semiconductor layer **202a** and the semiconductor layer **208a** are integrally formed. On the contrary, in an embodiment in which the semiconductor layer **202a** and the semiconductor layer **208a** have different materials, an interface may exist between the semiconductor layer **202a** and the semiconductor layer **208a**.

Referring to FIG. 2F and FIG. 3F, portions of the dielectric layer **200** are removed, to form trenches **216A**, **216B** and **216C**. In some embodiments, with the patterned photoresist layer **210**, the dielectric layer **200** is partially removed through an etching process. The etching process includes, for example, an isotropic etching process such as wet etch or an anisotropic etching process such as dry etch. In some embodiments, an etchant for the wet etch includes a combination of hydrogen fluoride (HF) and ammonia (NH₃), a combination of HF and tetramethylammonium hydroxide (TMAH), or the like. On the other hand, the dry etch process includes, for example, reactive ion etch (RIE), inductively coupled plasma (ICP) etch, electron cyclotron resonance (ECR) etch, neutral beam etch (NBE), and/or the like. As mentioned above, the etching selectivity between the semiconductor layer **202a** and the dielectric layer **200** and between the semiconductor layer **208a** and the dielectric layer **200** is high. Therefore, during the etching process, the etchant may selectively remove the dielectric layer **200** without damaging the exposed semiconductor layer **212A**, semiconductor layer **212B** and semiconductor nanosheet

214. Then, the patterned photoresist layer **210** is removed by a removal process such as an ashing process.

In some embodiments, the trench **216A** is formed between the source electrodes **206A** and the drain electrodes **206B** which are paired in the same row R, the trench **216B** is formed between different rows R (e.g., between different pairs of source and drain electrodes **206A** and **206B** in different rows R), and the trench **216C** is formed between the columns C (e.g., between different pairs of source and drain electrodes **206A** and **206B** in the same row R). The trench **216A** is disposed below the semiconductor nanosheets **214**. For example, as shown in FIG. 3F, sidewalls **217s** of the trench **216A** are substantially flush with adjacent sidewalls **207s** of a pair of the source and drain electrodes **206A** and **206B**. Accordingly, the semiconductor nanosheet **214** may be suspended over the dielectric layer **200** between a pair of source and drain electrodes **206A** and **206B**. In some embodiments, the trenches **216A** and the trenches **216B** are extended along the direction D2, and the trenches **216A** and the trenches **216B** are alternately arranged along the direction D1. The trenches **216C** may be extended along the direction D1 and arranged along the direction D2. The trenches **216C** cross over the trenches **216A** and the trenches **216B**, and the trenches **216A**, **216B** and **216C** are connected to form a net-shaped trench, for example. In some embodiments, a depth of the trenches **216A**, **216B** and **216C** is substantially the same. For example, the depth of the trenches **216A**, **216B** and **216C** is in a range of 10 nm to 100 nm.

Referring to FIG. 2G and FIG. 3G, a memory material **218** and a conductive material **220** may be sequentially formed. For example, the memory material **218** is formed on the exposed surfaces of the semiconductor nanosheet **214** and the trenches **216A** to **216C**. The memory material **218** may be deposited conformally on the exposed surfaces of the semiconductor nanosheet **214** and bottom and sidewall surfaces of the trenches **216A** to **216C**. In some embodiments, the memory material **218** includes materials that are capable of switching between two different polarization directions by applying an appropriate voltage differential across the memory material **218**. For example, the memory material **218** includes ferroelectric material or the like.

The memory material **218** may include $\text{Hf}_{1-x}\text{Zr}_x\text{O}_2$ (HfO₂), $\text{Hf}_{1-x}\text{Er}_x\text{O}$, $\text{Hf}_{1-x}\text{La}_x\text{O}$, $\text{Hf}_{1-x}\text{Y}_x\text{O}$, $\text{Hf}_{1-x}\text{Gd}_x\text{O}$, $\text{Hf}_{1-x}\text{Al}_x\text{O}$, $\text{Hf}_{1-x}\text{Ti}_x\text{O}$, $\text{Hf}_{1-x}\text{Ta}_x\text{O}$, BaMgF_4 , BaTiO_3 , PbZrO_3 , $(\text{Ba}, \text{Sr})\text{TiO}_3$, $\text{Bi}_4\text{Ti}_3\text{O}_{12}$, LiNbO_3 , LiTaO_3 , $(\text{Pb}, \text{La})\text{TiO}_3$ such as PbTiO_3 , $\text{Pb}(\text{Zr}, \text{Ti})\text{O}_3$, $(\text{Pb}, \text{La})(\text{Zr}, \text{Ti})\text{O}_3$ such as PbZrO_3 , $\text{SrBi}_2\text{Ta}_2\text{O}_9$, $\text{Bi}_{4-x}\text{La}_x\text{Ti}_3\text{O}_{12}$ (BLT), BiFeO_3 , YMnO_3 , YbMnO_3 , BiMnO_3 , $\text{Pb}(\text{Fe}_{0.5}\text{W}_{0.5})_3$, LiNbO_3 , NaNbO_3 , KNbO_3 , KTaO_3 , BiScO_3 , BiFeO_3 , the like or a combination thereof. In some embodiments, the memory material **218** includes different ferroelectric materials or different types of memory materials. The memory material **218** may be formed by suitable fabrication techniques such as ALD, CVD, metalorganic CVD (MOCVD), PVD, thermal oxidation, UV-ozone oxidation, remote plasma atomic layer deposition (RPALD), plasma-enhanced atomic layer deposition (PEALD), molecular beam deposition (MBD), or combinations thereof.

In some embodiments, the memory material **218** has a thickness of about 1-20 nm, such as 5-10 nm. Other thickness ranges (e.g., more than 20 nm or 5-15 nm) may be applicable. In some embodiments, the memory material **218** is formed in a fully amorphous state. In alternative embodiments, the memory material **218** is formed in a partially crystalline state; that is, the memory material **218** is formed in a mixed crystalline-amorphous state and having some

degree of structural order. In alternative embodiments, the memory material **218** is formed in a fully crystalline state. In some embodiments, the memory material **218** is a single layer. In alternative embodiments, the memory material **218** is a multi-layer structure. After the memory material **218** is deposited, an annealing step may be performed, so as to achieve a desired crystalline lattice structure for the memory material **218**. In some embodiments, upon the annealing process, the memory material **218** is transformed from an amorphous state to a partially or fully crystalline state. In alternative embodiments, upon the annealing, the memory material **218** is transformed from a partially crystalline state to a fully crystalline state.

In alternative embodiments, the memory material **218** is replaced with a gate dielectric material. In such embodiments, the gate dielectric material includes silicon oxide, silicon nitride, silicon oxynitride, high-k dielectrics, or a combination thereof. It should be noted that the high-k dielectric materials are generally dielectric materials having a dielectric constant higher than 4, greater than about 12, greater than about 16, or even greater than about 20. The gate dielectric material may include metal oxides, metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates, oxynitrides of metals, metal aluminates, or combinations thereof.

After forming the memory material **218**, the conductive material **220** may be formed to wrap the semiconductor nanosheets **214** and fills space above the semiconductor nanosheets **214** between the source and drain electrodes **206A** and **206B** and the trenches **216A** to **216C**. In some embodiments, the conductive material **220** includes Mo, Ti, Pd, W, Co, Cr, Cu, Ni, Ta, Pt, Au, Al, TiW, TiN, TaN, WN, WCN, any other suitable metal-containing material, or a combination thereof. The conductive material **220** may be formed by a deposition process such as ALD, CVD, PVD, or the like. In some embodiments, a barrier layer (not shown) is optionally formed between the conductive material **220** and the memory material **218**, so as to avoid diffusion of atoms between elements. In some embodiments, the barrier layer includes TiN, TaN, TiSiN, TaSiN, WSiN, TiC, TaC, TiAlC, TaAlC, TiAlN, TaAlN, or a combination thereof.

The memory material **218** and the conductive material **220** may further formed outside the trenches **216A** to **216C** such as formed on the semiconductor layers **212A** and **212B**. Thus, after forming the memory material **218** and the conductive material **220**, a planarization process (e.g., a CMP, etch back, or the like) may be performed to remove excess portions of the memory material **218** and the conductive material **220**. For example, by using the semiconductor layers **212A** and **212B** as polish stop layers or etch stop layers, the memory material **218** and the conductive material **220** higher than surfaces (e.g., top surfaces) of the semiconductor layers **212A** and **212B** are removed. Thus, as shown in FIG. 2G and FIG. 3G, after removal, surfaces (e.g., top surfaces) of the memory material **218** and the conductive material **220** are substantially coplanar with surfaces (e.g., top surfaces) of the semiconductor layers **212A** and **212B**. In some embodiments, as shown in FIG. 2G, the conductive material **220** is net-shaped.

Referring to FIG. 2H and FIG. 3H, portions of the conductive material **220** in the trenches **216B** are replaced with dielectric layers **226**. In some embodiments, portions of the conductive material **220** in the trenches **216B** are removed through an etching process by using a mask (not shown). For example, the conductive material **220** in the trenches **216B** is exposed by the mask, and then the exposed

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conductive material **220** is removed by an anisotropic etching process such as dry etch or an isotropic etching process such as wet etch. After the removal, the dielectric layers **226** are formed to fill the trenches **216B** respectively, for example. In some embodiments, the dielectric layer **226** is in direct contact with the memory layer **222** and is surrounded by the memory layer **222**. In some embodiments, as shown in FIG. 2H, the dielectric layer **226** has an inverted T shape. For example, a first portion of the dielectric layer **226** surrounded by the dielectric layer **200** has a first width, and a second portion of the dielectric layer **226** between the source and drain electrodes **206A** and **206B** has a second width smaller than the first width. In some embodiments, the dielectric layer **226** continuously extends between the adjacent source and drain electrodes **206A** and **206B** in different rows R. For example, the dielectric layer **226** continuously extends between the source electrodes **206A** in the first row R and the drain electrode **206B** in the second row R which are immediately adjacent to each other. The dielectric layer **226** may extend in the direction D2 substantially parallel to the extending direction (e.g., the direction D2) of the gate electrode **224**. The dielectric layers **226** and the gate electrodes **224** are alternately arranged along the direction D1, for example. The dielectric layers **226** may include silicon oxide, silicon nitride, silicon oxynitride, the like or a combination thereof, and dielectric layers **226** may be deposited by CVD, PVD, ALD, PECVD, the like or a combination thereof. Optionally, after deposition, a planarization process (e.g., a CMP, etch back, or the like) may be performed to remove excess portions of the dielectric materials. In some embodiments, the material of the dielectric layers **226** is substantially the same as the dielectric layer **200**. In such embodiments, an interface may not exist between the dielectric layer **200** and the dielectric layers **226**, and the dielectric layer **200** and the dielectric layers **226** are integrally formed. However, the disclosure is not limited thereto. The material of the dielectric layers **226** may be different from the dielectric layer **200**. In some embodiments, from a top view, the dielectric layer **226** is a straight line.

After forming the dielectric layers **226**, the remained memory material **218** in the trenches **216A**, **216B** and **216C** and the remained conductive material **220** in the trenches **216A** and **216C** are referred to as a memory layer **222** and a gate electrode **224**, respectively. As shown in FIG. 3H and FIG. 4A, the gate electrode **224** wraps around the semiconductor nanosheet **214** and is disposed between the source and drain electrodes **206A** and **206B**, for example. In some embodiments, as shown in FIG. 2H, the gate electrode **224** has an inverted T shape. For example, a first portion of the gate electrode **224** (e.g., below the semiconductor nanosheet **214**) surrounded by the dielectric layer **200** has a first width, and a second portion of the gate electrode **224** (e.g., above the semiconductor nanosheet **214**) between the source and drain electrodes **206A** and **206B** has a second width smaller than the first width. In some embodiments, the gate electrode **224** continuously extends between the source and drain electrodes **206A** and **206B** of plural pairs in the same row R. For example, the gate electrode **224** is disposed between the source electrodes **206A** and the drain electrodes **206B** which are paired in the same row R. In addition, the gate electrode **224** may further extend between the source electrodes **206A** and/or between the drain electrodes **206B** in the same row R. In such embodiments, the gate electrode **224** is shaped as a net and has a cross shape at an intersection **228A** of the source and drain electrodes **206A** and **206B** in the same row R. In some embodiments, as shown in FIG. 2H, an outer sidewall of the gate electrode **224** is substantially flush with

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an inner sidewall of the memory layer **222** surrounding the dielectric layer **226**. In some embodiments, the dielectric layer **226** is electrically isolated the gate electrodes **224** in different rows R. The dielectric layer **226** is in direct contact with the gate electrode **224**, for example.

In some embodiments, the memory layer **222** is continuously disposed along with the gate electrode **224**. The memory layer **222** is disposed between the gate electrode **224** and the semiconductor nanosheet **214**, between the gate electrode **224** and the semiconductor layer **212A**, between the gate electrode **224** and the semiconductor layer **212B** and between the gate electrode **224** and the dielectric layer **200**, for example. In some embodiments, the memory layer **222** is in direct contact with the gate electrode **224**, the semiconductor nanosheet **214**, the semiconductor layer **212A**, the semiconductor layer **212B** and the dielectric layer **200**. In some embodiments, the memory layer **222** is further disposed between and in direct contact with the dielectric layer **200** and the dielectric layer **226**. For example, the memory layer **222** surrounds the dielectric layer **226**. In some embodiments, the memory layer **222** is also referred to as a gate dielectric layer.

In the resulting structure, first surfaces (e.g., top surfaces) **223a**, **225a**, **213a**, **227a** of the memory layer **222**, the gate electrode **224**, the semiconductor layers **212A**, **212B** (i.e., the semiconductor layers **208a** of the semiconductor layers **212A**, **212B**) and the dielectric layer **226** may be substantially coplanar (e.g., within process variations). Second surfaces (e.g., bottom surfaces) **223b**, **225b**, **213b**, **227b** of the memory layer **222**, the gate electrode **224**, the semiconductor layers **212A**, **212B** (i.e., the semiconductor layers **202a** of the semiconductor layers **212A**, **212B**) and the dielectric layer **226** are disposed opposite to the first surfaces **223a**, **225a**, **213a**, **227a**. The second surface **225b** of the gate electrode **224** is substantially coplanar with the second surface **227b** of the dielectric layer **226**, and the second surface **223b** of the memory layer **222** below the gate electrode **224** is substantially coplanar with the second surface **223b** of the memory layer **222** below the dielectric layer **226**, for example. The second surfaces **213b** of the semiconductor layers **212A**, **212B** (i.e., the semiconductor layers **202a** of the semiconductor layers **212A**, **212B**) may be substantially coplanar with a surface **215a** (e.g., bottom surface) of the semiconductor nanosheet **214**, and may be disposed between the first and second surfaces **223a** and **223b** of the memory layer **222** and between the first and second surfaces **225a** and **225b** of the gate electrode **224**.

In some embodiments, the cross-sectional view of the semiconductor nanosheet **214** (i.e., channel layer) is circular (as shown in FIG. 4B). However, the disclosure is not limited thereto. The semiconductor nanosheet **214** may have any suitable cross-sectional view such as oval (as shown in FIG. 5A), square (as shown in FIG. 5B) and rectangular (as shown in FIG. 5C). In some embodiments, the memory layer **222** is illustrated as a single layer as shown in FIG. 4B to FIG. 5C. However, the disclosure is not limited thereto. In alternative embodiments, as shown in FIG. 5D and FIG. 5E, the memory layer **222** is a multi-layer structure. For example, the memory layer **222** includes a plurality of sublayers **222a**, **222b**, **222c** between the semiconductor nanosheet **214** and the gate electrode **224**. The materials of the sublayers **222a**, **222b**, **222c** may be selected based on the interfacial property to the semiconductor nanosheet **214** or the gate electrode **224**. For example, the sublayer **222a** is in direct contact with the semiconductor nanosheet **214**, and thus it is suitable for the sublayer **222a** to be electrically compatible with the semiconductor nanosheet **214**. Simi-

larly, the outermost sublayer (e.g., sublayer **222b** or sublayer **222c**) is in direct contact with the gate electrode **224**, and thus it is suitable for the outermost sublayer (e.g., sublayer **222b** or sublayer **222c**) to be electrically compatible with the gate electrode **224**. The sublayers **222a**, **222b**, **222c** may have different materials or the same material with different dopants or different dopant concentration. For example, the sublayers **222a**, **222b**, **222c** include $\text{Hf}_{1-x}\text{Zr}_x\text{O}_2$ (HZO)-based material, and ratios of Hf and Zr of the sublayers **222a**, **222b**, **222c** are different from each other.

In some embodiments, after formation of the dielectric layer **226**, a memory device is formed. The memory device includes a memory array including a plurality of memory cells MC arranged in a plurality of rows R and a plurality of columns C. As shown in FIG. 2H and FIG. 3H, it is seen that each memory cell MC includes a GAA transistor (e.g., a GAA ferroelectric TFT) T with the memory layer **222**. In alternative embodiments, the memory layer **222** is replaced with a gate dielectric layer including a dielectric material such as silicon oxide, silicon nitride, silicon oxynitride, and high-k dielectrics. In some embodiments, each GAA transistor T of the memory cell MC includes the gate electrode **224** (e.g., electrically connected to a word line), the source and drain electrodes **206A** and **206B** (e.g., electrically connected to a source line and a bit line) and the semiconductor nanosheet **214** functioning as the channel layer. The dielectric layers **200** and **226** are used for isolating memory cells MC. In some embodiments, the memory cells MC are disposed on the dielectric layer **200** and surrounded by the dielectric layer **226**. The dielectric layers **200** and **226** may be collectively referred to as a dielectric structure, and thus as shown in FIG. 2H and FIG. 3H, the memory cells MC are embedded in the dielectric structure.

In some embodiments, the memory layer **222** is used to store the digital information (e.g., a bit "1" or "0") stored in the memory cell MC. In some embodiments, the GAA ferroelectric TFT is integrated into CMOS BEOL process for computing-in-memory application due to its low-temperature process. Furthermore, the GAA ferroelectric TFT may show improved electrical properties such as gate control ability, low leakage current, low resistance and high on/off current ratio.

Referring to FIG. 2I and FIG. 3I, conductive lines **230A** and conductive lines **230B** are formed to electrically connect the source electrodes **206A** and the drain electrodes **206B**, respectively. In some embodiments, the conductive lines **230A** are formed above the source electrodes **206A**, and the conductive lines **230A** are electrically connected to the source electrodes **206A** through a plurality of conductive contacts **232A** therebetween. The conductive lines **230B** are formed above the drain electrodes **206B**, and the conductive lines **230B** are electrically connected to the drain electrodes **206B** through a plurality of conductive contacts **232B** therebetween. For example, the conductive contacts **232A** penetrate the semiconductor layer **212A** to electrically connect to the source electrodes **206A**, and similarly, the conductive contacts **232B** penetrate the semiconductor layer **212B** to electrically connect to the drain electrodes **206B**. In some embodiments, the conductive lines **230A**, **230B** are formed using a combination of photolithography and etching techniques. The conductive lines **230A**, **230B** may include copper, silver, gold, tungsten, aluminum, combinations thereof, or the like. In addition, the conductive lines **230A**, **230B** may have other configurations.

The conductive lines **230A** and the conductive lines **230B** may each extend in the direction D2 perpendicular to the extending direction (e.g., the direction D1) of the gate

electrodes **224**. The conductive lines **230A** and the conductive lines **230B** may be parallel to each other and alternately arranged over the dielectric layer **200** along the direction D1. In some embodiments, the conductive lines **230A**, **230B** and the gate electrodes **224** connect the memory device to an underlying/overlying circuitry (e.g., control circuitry) and/or signal, power, and ground lines, respectively. Other conductive contacts or vias may be formed to electrically connect the conductive lines **230A**, **230B** and the gate electrodes **224** to the underlying active devices of the substrate. In alternative embodiments, routing and/or power lines to and from the memory device are provided by an interconnect structure formed over the memory device.

In some embodiments, as shown in FIG. 2H, the dielectric layers **226** are merely formed in the trenches **216B**. However, the disclosure is not limited thereto. In alternative embodiments, the dielectric layers **226** further extend into the trenches **216C**. In such embodiments, during the step of removal of the conductive material **220** in the trenches **216B**, the conductive material **220** in portions of the trenches **216C** are also removed. Thus, as shown in FIG. 6, the dielectric layer **226** has a cross shape at an intersection **228B** of the source and drain electrodes **206A** and **206B** in different rows R. The dielectric layer **226** may further extend between the adjacent source and electrodes **206A** and **206B** in different rows R. In some embodiments, as shown in FIG. 6, both outer sidewalls of the dielectric layer **226** at the intersection **228B** extend beyond the memory layer **222**. In such embodiments, the dielectric layer **226** and the gate electrode **224** are both shaped as a net. However, the disclosure is not limited thereto. In alternative embodiments, only a first outer sidewall of the dielectric layer **226** extends beyond the memory layer **222** while a second outer sidewall opposite to the first outer sidewall of the dielectric layer **226** remains being substantially flush with the memory layer **222**. In an embodiment (not shown), the dielectric layer **226** fills the trench **216C** entirely, and thus the dielectric layer **226** has a net shape while the gate electrode **224** is a straight line.

In some embodiments, as mentioned before with reference to FIG. 2G and FIG. 3G, since the semiconductor layers **212A** and **212B** is used as polish stop layers or etch stop layers, surfaces (e.g., top surfaces) of the memory material **218** and the conductive material **220** are substantially coplanar with surfaces (e.g., top surfaces) of the semiconductor layers **212A** and **212B**. Accordingly, the formed memory layer **222** and gate electrode **224** also have the surfaces (e.g., top surfaces) **223a**, **225a** substantially coplanar with surfaces (e.g., top surfaces) **213a** of the semiconductor layers **212A** and **212B** as shown in FIG. 2H and FIG. 3H. However, the disclosure is not limited thereto. In alternative embodiments, during the partial removal of the conductive material **220**, the memory material **218** may serve as a polish stop layer or an etch stop layer, and thus the memory material **218** remains on the semiconductor layers **212A** and **212B** without removing. Accordingly, as shown in FIG. 7A and FIG. 7B, the formed memory layer **222** further extends onto the surfaces **213a** (e.g., top surfaces) of the semiconductor layers **212A** and **212B**. In such embodiments, the memory layer **222** surrounds the dielectric layer **226**, extends on the semiconductor layers **212A** and **212B** and wraps the semiconductor nanosheet **214** continuously. The surface **223a** of the memory layer **222** is substantially flush with the surfaces **227a**, **225a** of the dielectric layer **226** and the gate electrode **224**, for example. Furthermore, in such embodiments, as shown in FIG. 7A and FIG. 7B, the conductive contacts **232A** and **232B** penetrates the memory

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layer 222 and the semiconductor layer 212A to electrically connect to the source and drain electrodes 206A and 206B respectively.

FIG. 8 illustrates a manufacturing method of a semiconductor package in accordance with some embodiments. Although the method is illustrated and/or described as a series of acts or events, it will be appreciated that the method is not limited to the illustrated ordering or acts. Thus, in some embodiments, the acts may be carried out in different orders than illustrated, and/or may be carried out concurrently. Further, in some embodiments, the illustrated acts or events may be subdivided into multiple acts or events, which may be carried out at separate times or concurrently with other acts or sub-acts. In some embodiments, some illustrated acts or events may be omitted, and other un-illustrated acts or events may be included.

At act S300, a first semiconductor material is formed over a first dielectric layer. FIG. 2A and FIG. 3A illustrate views corresponding to some embodiments of act S300.

At act S302, a first conductive material on the first semiconductor material. FIG. 2A and FIG. 3A illustrate views corresponding to some embodiments of act S302.

At act S304, the first conductive material is patterned to form plural pairs of source and drain electrodes separated from each other. FIG. 2B and FIG. 3B illustrate views corresponding to some embodiments of act S304.

At act S306, a second semiconductor material is formed on the plural pairs of source and drain electrodes and the first semiconductor material. FIG. 2C and FIG. 3C illustrate views corresponding to some embodiments of act S306.

At act S308, portions of the first semiconductor material, the second semiconductor material and the first dielectric layer respectively between each pair of source and drain electrodes are removed, to form a plurality of semiconductor nanosheets respectively between each pair of source and drain electrodes and a plurality of first trenches respectively below the semiconductor nanosheets. FIG. 2D to FIG. 2F and FIG. 3D to FIG. 3F illustrate views corresponding to some embodiments of act S306.

At act S310, a memory layer and a gate electrode are formed to wrap around the semiconductor nanosheets. FIG. 2G, FIG. 2H, FIG. 3G, FIG. 3H, FIG. 6, FIG. 7A and FIG. 7B illustrate views corresponding to some embodiments of act S310.

In accordance with some embodiments of the disclosure, a transistor includes a first semiconductor layer, a second semiconductor layer, a semiconductor nanosheet, a gate electrode and source and drain electrodes. The semiconductor nanosheet is physically connected to the first semiconductor layer and the second semiconductor layer. The gate electrode wraps around the semiconductor nanosheet. The source and drain electrodes are disposed at opposite sides of the gate electrode. The first semiconductor layer surrounds the source electrode, the second semiconductor layer surrounds the drain electrode, and the semiconductor nanosheet is disposed between the source and drain electrodes.

In accordance with some embodiments of the disclosure, a memory device includes a plurality of memory cells. The memory cells include plural pairs of source and drain electrodes, a plurality of semiconductor nanosheets, a gate electrode and a memory layer. The source and drain electrodes are separated from each other. Each semiconductor nanosheet is disposed between each pair of source and drain electrodes. The gate electrode continuously wraps around the semiconductor nanosheets, wherein the gate electrode is continuously disposed between the plural pairs of source and

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drain electrodes. The memory layer is disposed between the gate electrode and the semiconductor nanosheets.

In accordance with some embodiments of the disclosure, a manufacturing method of a memory device includes at least the following steps. A first semiconductor material is formed over a first dielectric layer. A first conductive material is formed on the first semiconductor material. The first conductive material is patterned to form plural pairs of source and drain electrodes separated from each other. A second semiconductor material is formed on the plural pairs of source and drain electrodes and the first semiconductor material. Portions of the first semiconductor material, the second semiconductor material and the first dielectric layer respectively between each pair of source and drain electrodes are removed, to form a plurality of semiconductor nanosheets respectively between each pair of source and drain electrodes and a plurality of first trenches respectively below the semiconductor nanosheets. A memory layer and a gate electrode are formed to wrap around the semiconductor nanosheets.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A transistor, comprising:

a first semiconductor layer, a second semiconductor layer and a semiconductor nanosheet physically connected to the first semiconductor layer and the second semiconductor layer;

a gate electrode, wrapping around the semiconductor nanosheet; and

source and drain electrodes at opposite sides of the gate electrode, wherein the first semiconductor layer surrounds the source electrode, the second semiconductor layer surrounds the drain electrode, and the semiconductor nanosheet is disposed between the source and drain electrodes, wherein the first semiconductor layer is disposed on opposite surfaces and sidewall surfaces of the source electrode, and the second semiconductor layer is disposed on opposite surfaces and sidewall surfaces of the drain electrode.

2. The transistor of claim 1, further comprising a memory layer between the gate electrode and the semiconductor nanosheet, wherein a surface of the gate electrode is substantially coplanar with surfaces of the memory layer, the first semiconductor layer and the second semiconductor layer.

3. The transistor of claim 2, further comprising a dielectric layer surrounded by the memory layer, wherein a surface of the dielectric layer is substantially coplanar with the surfaces of the memory layer, the first semiconductor layer, the second semiconductor layer and the gate electrode.

4. The transistor of claim 1, further comprising a memory layer between the gate electrode and the semiconductor nanosheet and a dielectric layer surrounded by the memory

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layer, wherein a surface of the dielectric layer is substantially coplanar with surfaces of the memory layer and the gate electrode.

5. The transistor of claim 4, wherein the first semiconductor layer is disposed between the memory layer and the source electrode, and the second semiconductor layer is disposed between the memory layer and the drain electrode.

6. The transistor of claim 1, further comprising a dielectric layer below the source and drain electrodes, wherein the gate electrode has a first width in the dielectric layer and a second width between the source and drain electrodes, and the second width is smaller than the first width.

7. A memory device, comprising:

a plurality of memory cells, comprising:

plural pairs of source and drain electrodes separated from each other;

a plurality of semiconductor nanosheets, wherein each semiconductor nanosheet is disposed between each pair of source and drain electrodes;

a gate electrode, continuously wrapping around the semiconductor nanosheets, wherein the gate electrode is continuously disposed between the plural pairs of source and drain electrodes; and

a memory layer, between the gate electrode and the semiconductor nanosheets, wherein the gate electrode has a cross shape at an intersection of the plural pairs of source and drain electrodes.

8. The memory device of claim 7, further comprising a plurality of semiconductor layers respectively surrounding the plural pairs of source and drain electrodes, wherein the semiconductor layers are physically connected to the semiconductor nanosheets, respectively.

9. The memory device of claim 8, wherein a surface of the gate electrode is substantially coplanar with surfaces of the memory layer and the semiconductor layers.

10. The memory device of claim 7, further comprising a first dielectric layer and a second dielectric layer on the first dielectric layer, wherein the memory cells are disposed on the first dielectric layer, and a first surface of the second dielectric layer is substantially coplanar with first surfaces of the memory layer and the gate electrode.

11. The memory device of claim 10, wherein the memory layer is further disposed between the first dielectric layer and the second dielectric layer.

12. The memory device of claim 10, wherein the gate electrode and the second dielectric layer are extended along a first direction respectively, and the gate electrode and the second dielectric layer are arranged along a second direction substantially perpendicular to the first direction.

13. The memory device of claim 10, wherein the second dielectric layer is in direct contact with the gate electrode.

14. The memory device of claim 10, wherein the second dielectric layer has a cross shape at an intersection of the plural pairs of source and drain electrodes.

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15. The transistor of claim 7, wherein the memory layer includes a ferroelectric material.

16. A manufacturing method of a memory device, comprising:

forming a first semiconductor material over a first dielectric layer;

forming a first conductive material on the first semiconductor material;

patterning the first conductive material to form plural pairs of source and drain electrodes separated from each other;

forming a second semiconductor material on the plural pairs of source and drain electrodes and the first semiconductor material;

removing portions of the first semiconductor material, the second semiconductor material and the first dielectric layer respectively between each pair of source and drain electrodes, to form a plurality of semiconductor nanosheets respectively between each pair of source and drain electrodes and a plurality of first trenches respectively below the semiconductor nanosheets; and forming a memory layer and a gate electrode to wrap around the semiconductor nanosheets.

17. The method of claim 16, wherein removing the portions of the first semiconductor material, the second semiconductor material and the first dielectric layer between each pair of source and drain electrodes comprises:

removing the first semiconductor material, the second semiconductor material and the first dielectric layer respectively disposed between the source and drain electrodes of each pair, to form the semiconductor nanosheets and the first trenches; and

removing the first semiconductor material, the second semiconductor material and the first dielectric layer between the plural pairs of source and drain electrodes, to form second trenches, wherein the first trenches and the second trenches are alternately arranged.

18. The method of claim 17, wherein the first semiconductor material, the second semiconductor material and the first dielectric layer between the plural pairs of source and drain electrodes are entirely removed.

19. The method of claim 17, wherein forming the memory layer and the gate electrode comprises:

forming a memory material over the semiconductor nanosheets and the plural pairs of source and drain electrodes; and

forming a conductive material over the memory material to wrap the semiconductor nanosheets and fill the first trenches.

20. The method of claim 19, wherein the memory material and the conductive material further form in the second trenches, and the method further comprises replacing the conductive material in the second trenches with a dielectric material.

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